The documentation and process conversion measures necessary to comply with this revision shall be completed by 19 July 2021.

INCH-POUND

MIL-PRF-19500/391R
w/AMENDMENT 1
19 April 2021
SUPERSEDING
MIL-PRF-19500/391R
28 March 2019

PERFORMANCE SPECIFICATION SHEET

TRANSISTOR, NPN, SILICON, LOW-POWER, DEVICE TYPES 2N3019, 2N3057A, AND 2N3700,
ENCAPSULATED (THROUGH-HOLE AND SURFACE MOUNT) AND UNENCAPSULATED,
RADIATION HARDNESS ASSURANCE,
QUALITY LEVELS JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN, silicon, low-power transistors. Four levels of product assurance (JAN, JANTX, JANTXV and JANS) are provided for each device type as specified in MIL-PRF-19500. Two levels of product assurance (JANHC and JANKC) are provided for each unencapsulated device type. Provisions for radiation hardness assurance (RHA) to eight radiation levels is provided for quality levels JANTXV, JANS, JANHC, and JANKC.

1.2 Physical dimensions.

1.2.1 Package outlines. The device packages for the encapsulated device types are as follows: TO-205AA and TO-205AD (formerly TO-5 and TO-39) in accordance with figure 1, TO-206AB (formerlyTO-46) in accordance with figure 2, three terminal round metal can TO-206AA (formerly TO-18) in accordance with figure 3, and four terminal SMD package UB in accordance with figure 4.

* 1.2.2 Un-encapsulated die. The dimensions and topography for JANHC and JANKC unencapsulated die are as follows: The A version die (JANHCA and JANKCA) is in accordance with figure 5, the B version die (JANHCB and JANKCB) is in accordance with figure 6, the C version die (JANHCC and JANKCC) is in accordance with figure 7, and the D version die (JANHCD and JANKCD) is in accordance with figure 8.

1.3 Maximum ratings. Unless otherwise specified, $T_A$ = +25°C.

| Types | $I_C$ | $V_{CBO}$ | $V_{EBO}$ | $V_{CEO}$ | $T_J$ and $T_{STG}$ |
|---|---|---|---|---|---|
| | A dc | V dc | V dc | V dc | °C |
| All devices | 1 | 140 | 7 | 80 | -65 to +200 |

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1.3 Maximum ratings. Unless otherwise specified, $T_A$ = +25°C - Continued.

| Types | $P_T$ $T_A$ = +25°C (1) (2) | $P_T$ $T_C$ = +25°C (1) (2) | $P_T$ $T_{SP(IS)}$ = +25°C (1) (2) | $R_{\theta JA}$ (2) (3) (4) | $R_{\theta JC}$ (2) (3) | $R_{\theta JSP(IS)}$ (2) (3) |
|---|---|---|---|---|---|---|
| | W | W | W | °C/W | °C/W | °C/W |
| 2N3019 | 0.800 | 5 | N/A | 195 | 30 | N/A |
| 2N3019S | 0.800 | 5 | N/A | 195 | 30 | N/A |
| 2N3057A | 0.500 | 1.8 | N/A | 325 | 80 | N/A |
| 2N3700 | 0.500 | 1 | N/A | 325 | 150 | N/A |
| 2N3700UB | 0.500 | N/A | 1.5 | 325 | N/A | 90 |

* (1) For derating, see figures 9, 10, 11, 12, 13, and 14.
(2) See 3.3.
* (3) For thermal curves, see figures 15, 16, 17, 18, 19, 20, and 21.
* (4) For non-thermal conductive PCB or unknown PCB surface mount conditions in free air, substitute figures 9 and 13 for the UB package and use $R_{\theta JA}$.

1.4 Primary electrical characteristics.

| Limits | $h_{FE1}$ $V_{CE}$ = 10 V dc $I_C$ = 150 mA dc | $h_{FE2}$ $V_{CE}$ = 10 V dc $I_C$ = 0.1 mA dc | $h_{FE3}$ (1) $V_{CE}$ = 10 V dc $I_C$ = 10 mA dc | $h_{FE4}$ (1) $V_{CE}$ = 10 V dc $I_C$ = 500 mA dc |
|---|---|---|---|---|
| Min | 100 | 50 | 90 | 50 |
| Max | 300 | 300 | | 300 |

| Limit | $h_{FE5}$ (1) $V_{CE}$ = 10 V dc $I_C$ = 1 A dc | $\lvert h_{fe} \rvert$ f = 20 MHz $V_{CE}$ = 10 V dc $I_C$ = 50 mA dc | $C_{obo}$ 100 kHz ≤ f ≤ 1 MHz $V_{CB}$ = 10 V dc $I_E$ = 0 |
|---|---|---|---|
| | | | pF |
| Min | 15 | 5 | |
| Max | | 20 | 12 |

| Limits | $V_{CE(sat)1}$ (1) $I_C$ = 150 mA dc $I_B$ = 15 mA dc | $V_{CE(sat)2}$ (1) $I_C$ = 500 mA dc $I_B$ = 50 mA dc | $V_{BE(sat)}$ (1) $I_C$ = 150 mA dc $I_B$ = 15 mA dc |
|---|---|---|---|
| | V dc | V dc | V dc |
| Min | | | |
| Max | 0.2 | 0.5 | 1.1 |

(1) Pulsed, see 4.5.1.



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1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

1.5.1 JAN brand and quality level designators.

1.5.1.1 Encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JAN", "JANTX", "JANTXV" and "JANS".

1.5.1.2 Unencapsulated die. The quality level designators for unencapsulated die that are applicable for this specification sheet from the lowest to the highest level are as follows: "JANHC" and "JANKC".

1.5.2 RHA designator. The RHA levels applicable for this specification sheet from lowest to highest for quality levels JANS and JANKC are as follows: "M", "D", "P", "L", "R", "F", "G", and "H". The RHA levels applicable for this specification sheet from lowest to highest for JANTXV quality levels are as follows: "R" and "F".

1.5.3 Device type. The designation system for the device types of transistors covered by this specification sheet are as follows.

1.5.3.1 First number and first letter symbols. The transistors of this specification sheet use the first number and letter symbols "2N".

1.5.3.2 Second number symbols. The second number symbols for the transistors covered by this specification sheet are as follows: "3019", "3057", and "3700".

1.5.4 Suffix symbols. The following suffix letters are incorporated in the PIN as applicable.

1.5.4.1 Modified version designator. Device type 2N3057 is available only with an "A" suffix symbol. The "A" suffix symbol indicates an electrical parameter modified version of the device versus the non-A suffix device.

1.5.4.2 Package designators. The suffix symbols (or lack thereof) that designate the package outline for the devices covered by this specification sheet are as follows:

| | |
|---|---|
| | For device type 2N3019 only, a blank package designator identifies the package as a TO-205AA (see figure 1) with a dimension LL range from 1.50 minimum to 1.75 maximum. |
| S | For device type 2N3019 only, a "S" package designator identifies the package as a TO-205AD (see figure 1) with a dimension LL range from .500 minimum to .750 maximum. |
| | For device type 2N3057A only, a blank package designator identifies the package as a TO-206AB (see figure 2). |
| | For device type 2N3700 only, a blank package designator identifies the package as a TO-206AA (see figure 3). |
| UB | For device type 2N3700 only, a "UB" package designator identifies that the package is a 4-terminal metal lid (used as a shield and connected to fourth pad) SMD package (see figure 4). |

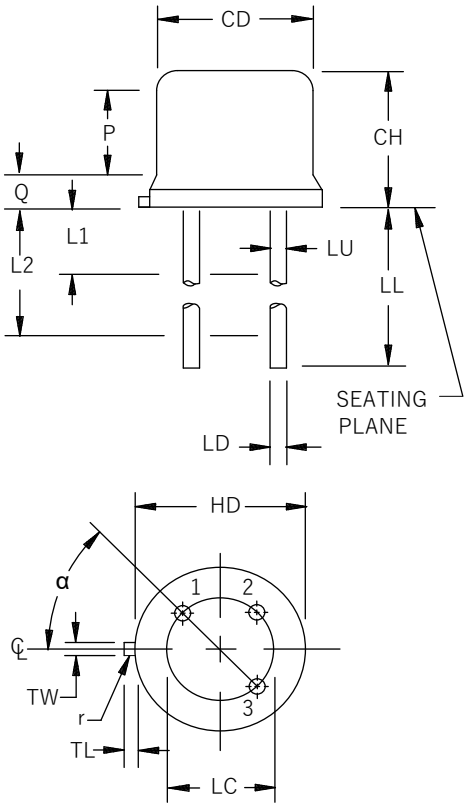
1.5.5 Lead finish. The lead finishes applicable to this specification sheet are listed on QML-19500.

* 1.5.6 Die identifiers for unencapsulated devices. The manufacturer die identifiers that are applicable for this specification sheet are "A" (see figure 5), "B" (see figure 6), "C" (see figure 7), and "D" (see figure 8).



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| Symbol | Dimensions | | | | Notes |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| | Min | Max | Min | Max | |
| CD | .305 | .335 | 7.75 | 8.51 | 4 |
| CH | .240 | .260 | 6.10 | 6.60 | |
| HD | .335 | .370 | 8.51 | 9.40 | 4 |
| LC | .200 TP | | 5.08 TP | | 5 |
| LD | .016 | .021 | 0.41 | 0.53 | 6, 7 |
| LL | 1.50 | 1.750 | 38.10 | 40.45 | 6, 7, 8 |
| LU | .016 | .019 | 0.41 | 0.48 | 6, 7 |
| L1 | | .050 | | 1.27 | 6, 7 |
| L2 | .250 | | 6.35 | | 6, 7 |
| P | .100 | | 2.54 | | |
| Q | | .050 | | 1.27 | 4 |
| TL | .029 | .045 | 0.74 | 1.14 | 9 |
| TW | .028 | .034 | 0.71 | 0.86 | 10 |
| r | | .010 | | 0.25 | 11 |
| α | 45° TP | | 45° TP | | 5 |




NOTES:

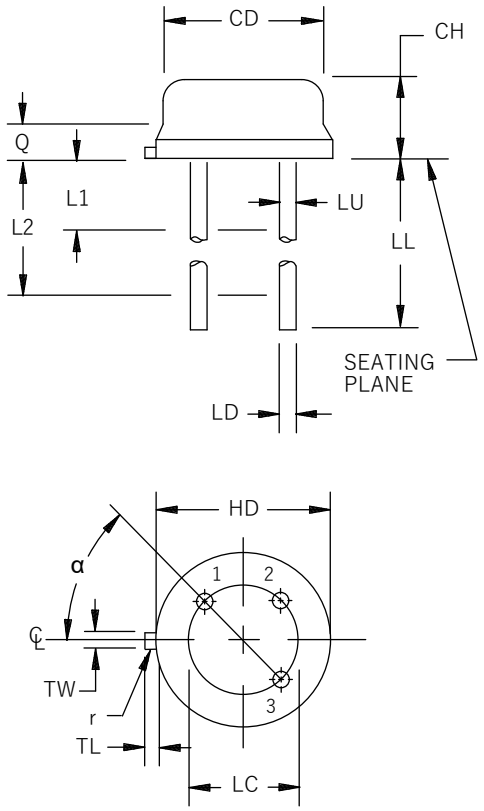
1. Dimension are in inches. Millimeters are given for general information only.
2. Terminal 1 = emitter, terminal 2 = base, terminal 3 = collector.
3. The collector shall be internally connected to the case.
4. Body contour optional within zone defined by dimensions CD, HD, and Q.
5. Leads at gauge plane .054 +.001 -.000 inch (1.37 +0.03 -0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods.
6. Dimension LU applies between dimensions L1 and L2. Dimension LD applies between dimensions L2 and LL minimum. Diameter is uncontrolled in dimension L1 and beyond dimension LL minimum.
7. All three leads.
8. For "S" suffix devices, dimension LL is .500 (12.70 mm) minimum, .750 (19.05 mm) maximum.
9. Dimension TL measured from maximum HD.
10. Beyond r (radius) maximum, dimension TW shall be held for a minimum length of .011 inch (0.28 mm).
11. Dimension r (radius) applies to both inside corners of tab.
* 12. In accordance with ASME Y14.5, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions for TO-205AA and AD packages (formerly TO-5) (device type 2N3019 only).



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| Symbol | Dimensions | | | | Notes |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| | Min | Max | Min | Max | |
| CD | .178 | .195 | 4.52 | 4.95 | |
| CH | .065 | .085 | 1.65 | 2.16 | |
| HD | .209 | .230 | 5.31 | 5.84 | |
| LC | .100 TP | | 2.54 TP | | 4 |
| LD | .016 | .021 | 0.41 | 0.53 | 5 |
| LL | .500 | 1.750 | 12.70 | 44.45 | 5, 6 |
| LU | .016 | .019 | 0.41 | 0.48 | 5, 6 |
| L1 | | .050 | | 1.27 | 5, 6 |
| L2 | .250 | | 6.35 | | 5, 6 |
| Q | | .040 | | 1.02 | |
| TL | .028 | .048 | 0.71 | 1.22 | 7 |
| TW | .036 | .046 | 0.91 | 1.17 | 8, 9 |
| r | | .010 | | 0.25 | 9, 10 |
| α | 45° TP | | 45° TP | | |



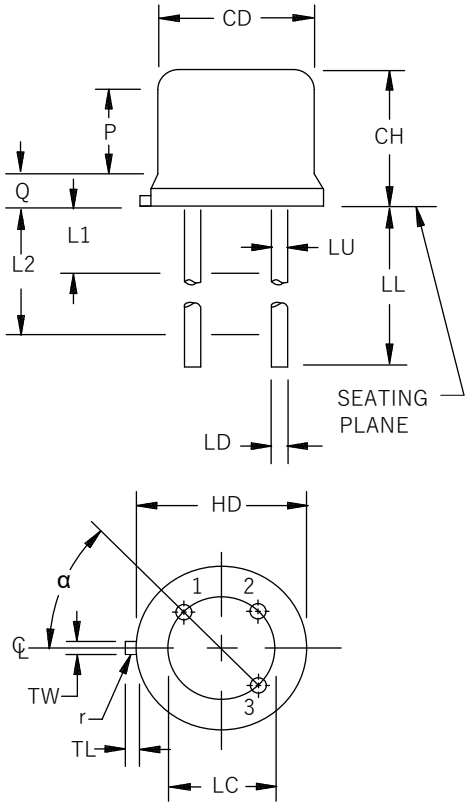

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Terminal 1 is emitter, terminal 2 is base, and terminal 3 is collector.
3. Terminal 3 (collector) is electrically connected to case.
4. Leads at gauge plane .054 inch (1.37 mm) +.001 inch (0.03 mm) -.000 inch (0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of TP relative to tab. Device may be measured by direct methods or by gauge.
5. All three leads.
6. Dimension LU applies between L1 and L2. Dimension LD applies between L2 and LL minimum.
7. Dimension TL is measured from dimension HD maximum.
8. Details of outline in this zone are optional.
9. Beyond r (radius) maximum, dimension TW shall be held for a minimum length of .011 inch (0.28 mm).
10. Dimension r (radius) applied to both inside corners of tab.
* 11. In accordance with ASME Y14.5, diameters are equivalent to ϕx symbol.

FIGURE 2. Physical dimensions for TO-206AB package (formerly TO-46) (device type 2N3057A only).



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| Symbol | Dimensions | | | | Note |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| | Min | Max | Min | Max | |
| CD | .178 | .195 | 4.52 | 4.95 | 4 |
| CH | .170 | .210 | 4.32 | 5.33 | |
| HD | .209 | .230 | 5.31 | 5.84 | 4 |
| LC | .100 TP | | 2.54 TP | | 5 |
| LD | .016 | .021 | 0.41 | 0.53 | 6, 7 |
| LL | .500 | .750 | 12.70 | 19.05 | 6, 7 |
| LU | .016 | .019 | 0.41 | 0.48 | 6, 7 |
| L1 | | .050 | | 1.27 | 6, 7 |
| L2 | .250 | | 6.35 | | 6, 7 |
| P | .100 | | 2.54 | | |
| Q | | .030 | | 0.76 | 4 |
| TL | .028 | .048 | 0.71 | 1.22 | 8 |
| TW | .036 | .046 | 0.91 | 1.17 | 9 |
| r | | .010 | | 0.25 | 10 |
| α | 45° TP | | 45° TP | | 5 |



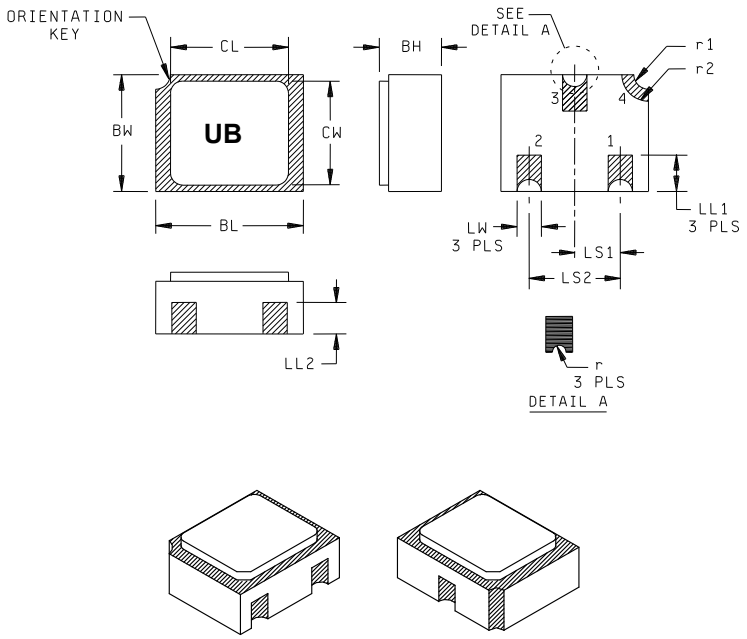

NOTES:

1. Dimension are in inches. Millimeters are given for general information only.
2. Terminal 1 = emitter, terminal 2 = base, terminal 3 = collector.
3. The collector shall be internally connected to the case.
4. Body contour optional within zone defined by dimensions CD, HD, and Q.
5. Leads at gauge plane .054 +.001 -.000 inch (1.37 +0.03 -0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods or by gauge
6. Dimension LU applies between dimensions L1 and L2. Dimension LD applies between dimensions L2 and LL minimum. Diameter is uncontrolled in dimension L1 and beyond dimension LL minimum.
7. All three leads.
8. Dimension TL measured from maximum HD.
9. Beyond r (radius) maximum, dimension TW shall be held for a minimum length of .011 inch (0.28 mm).
10. Dimension r (radius) applies to both inside corners of tab.
* 11. In accordance with ASME Y14.5, diameters are equivalent to ϕx symbology.

FIGURE 3. Physical dimensions for TO-206AA package (formerly TO-18) (device type 2N3700 only).



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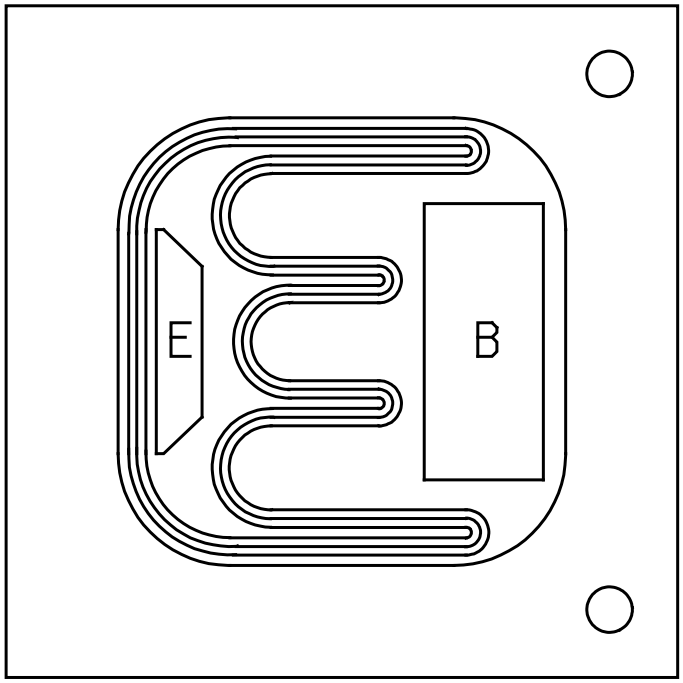

| Symbol | Dimensions | | | | Note |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| | Min | Max | Min | Max | |
| BH | .046 | .056 | 1.17 | 1.42 | |
| BL | .115 | .128 | 2.92 | 3.25 | |
| BW | .085 | .108 | 2.16 | 2.74 | |
| CL | | .128 | | 3.25 | |
| CW | | .108 | | 2.74 | |
| LL1 | .022 | .038 | 0.56 | 0.96 | |
| LL2 | .017 | .035 | 0.43 | 0.89 | |
| LS1 | .035 | .040 | 0.89 | 1.02 | |
| LS2 | .071 | .079 | 1.81 | 2.01 | |
| LW | .016 | .024 | 0.41 | 0.61 | |
| r | | .008 | | .203 | |
| r1 | | .012 | | .305 | |
| r2 | | .022 | | .559 | |

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Hatched areas on package denote metallized areas.
3. Pad 1 = Base, Pad 2 = Emitter, Pad 3 = Collector, Pad 4 = Shielding connected to the lid.
* 4. In accordance with ASME Y14.5, diameters are equivalent to ϕx symbology.

FIGURE 4. Physical dimensions for 4 terminal SMD packages (device type 2N3700UB only).



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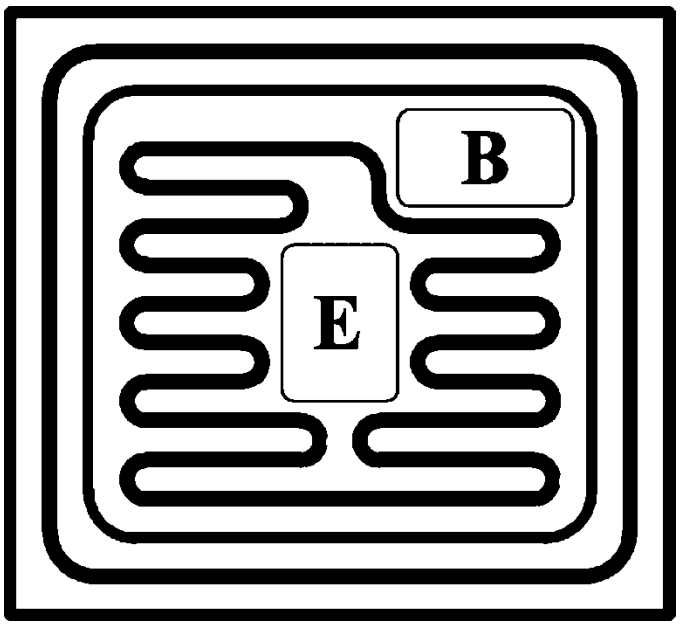

Physical characteristics (A-version):

1. Die size: .030 x .030 inch (0.762 x 0.762 mm).
2. Die thickness: .008 ±.0016 inch (0.2032 mm ±0.04064 mm).
3. Base pad: .004 x .010 inch (0.1016 mm x 0.254 mm).
4. Emitter pad: .0023 x .007 inch (0.05842 mm x 0.1778 mm).
5. Collector pad: Back side.
6. Top metal: Aluminum, 12,000 Å minimum; 14,500 Å nominal.
7. Backside metal: Gold, 6,500 ±1,950 Å.
8. Glassivation: $SiO_2$, 7,500 ±1,500 Å.

FIGURE 5. JANHCA and JANKCA die (2N3700 and 2N3019) dimensions.



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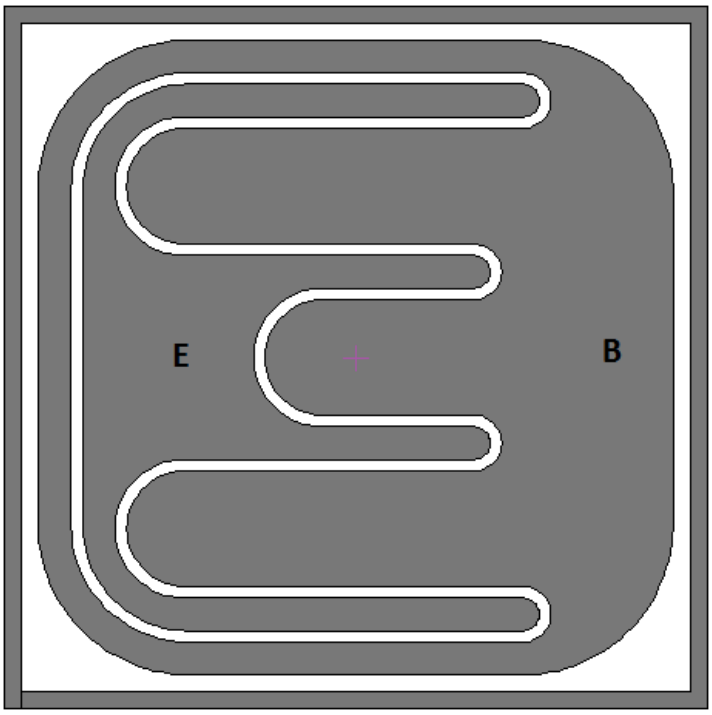

Physical characteristics (B-version):

| | | |
|---|---|---|
| 1. | Die size | .024 x .026 inch ±.002 inch (0.610 X 0.660 ±0.051 millimeter). |
| 2. | Die thickness | .010 inch ±.0015 inch (0.254 ±0.0381 millimeter). |
| 3. | Base pad | B = .004 inch x .006 inch (0.102 X 0.152 millimeter). |
| 4. | Emitter pad | E = .004 inch x .0055 inch (0.102 X 0.140 millimeter). |
| 5. | Collector pad. | Backside. |
| 6. | Top metal | Aluminum 15,000Å minimum, 18,000Å nominal. |
| 7. | Backside metal | Gold: 3,500Å minimum, 5,000Å nominal. |

FIGURE 6. JANHCB and JANKCB die (2N3700) dimensions.



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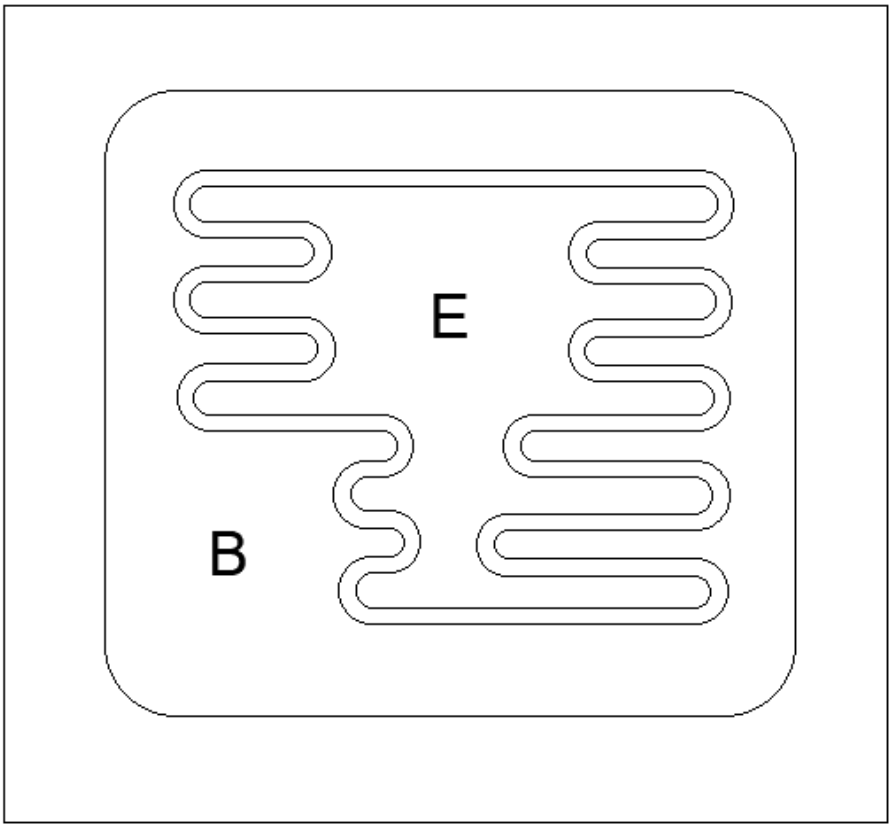

Physical characteristics (C-version):

| | | |
|---|---|---|
| 1. | Die size: | .023 x .023 inch ±.002 inch (0.584 X 0.584 ±0.051 millimeter). |
| 2. | Die thickness: | .010 inch ±.002 inch (0.254 ±0.508 millimeter). |
| 3. | Base pad: | B = .004 inch x .010 inch (0.102 X 0.254 millimeter). |
| 4. | Emitter pad: | E = .0039 inch x .0039 inch (0. 992 X 0.099 millimeter). |
| 5. | Collector pad: | Backside. |
| 6. | Top metal: | Aluminum 16,000Å minimum, 20,000Å nominal. |
| 7. | Backside metal: | Gold: 5,000Å ±500Å. |

FIGURE 7. JANHCC and JANKCC die (2N3700) dimensions.



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Physical characteristics (G-version):

| | | |
|---|---|---|
| 1. | Die Size: | .023 x .023 Inch [± .002 Inch] (0.584 mm x 0.584 mm [± 0.058mm]) |
| 2. | Die Thickness: | .008 Inch [± .0016 Inch] (0.203 mm [± 0.0406mm]) |
| 3. | Base pad: | .004 x .004 Inch (0.109 mm x 0.109 mm) |
| 4. | Emitter pad: | .004 x .004 Inch (0.109 mm x 0.109 mm |
| 5. | Collector: | Backside. |
| 6. | Top Metal: | Aluminum, 14,000 Å Minimum, 20,000 Å Nominal |
| 7. | Back Metal: | Gold, 8,400 Å Minimum, 12,000 Å Nominal |

* FIGURE 8. JANHCD and JANKCD die (2N3700 and2N3019) dimensions.



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## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML-19500) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

PCB ..... Printed circuit board.
$R_{\theta JA}$ ..... Thermal resistance junction to ambient.
$R_{\theta JC}$ ..... Thermal resistance junction to case.
$R_{\theta JSP(IS)}$..... Thermal resistance junction to solder pads (infinite sink mount to PCB).
$T_{SP(IS)}$..... Temperature of solder pads (infinite sink mount to PCB).
UB..... Surface mount case outlines (see figure 4).

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1, figure 2, figure 3, figure 4, figure 5, figure 6, figure 7 and figure 8 herein.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

* 3.4.2 Pin-out. The pin-out of the device shall be as shown on figures 1, 2, 3, and 4 as applicable.



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3.5 Marking.

3.5.1 Marking of all devices except "UB" suffix devices. Marking shall be in accordance with MIL-PRF-19500.

3.5.2 Marking of "UB" suffix devices. Marking on the UB package shall consist of an abbreviated PIN, the date code, and the manufacturers symbol or logo. The prefixes JAN, JANTX, JANTXV, and JANS can be abbreviated as J, JX, JV, and JS respectively. The "2N" prefix can also be omitted. The RHA designator (see 1.5.2) shall immediately precede (or replace) the device "2N" identifier (depending upon degree of abbreviation required).

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.8 RHA. Radiation hardness assurance requirements and test levels shall be as defined in MIL-PRF-19500.

3.9 Workmanship. Transistors shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I, II, and III).

d. Element evaluation (see 4.6).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table III tests, the tests specified in table III herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

4.2.2 JANHC and JANKC qualification. JANHC and JANKC qualification inspection shall be in accordance with MIL-PRF-19500.

4.2.3 Radiation hardened devices. See 4.4.4 herein and MIL-PRF-19500.



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4.3 Screening of encapsulated devices (quality levels JANTX, JANTXV, and JANS only). Screening shall be in accordance with table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen | Measurement | |
|---|---|---|
| | JANS | JANTX and JANTXV |
| (1) 3c | See 4.3.2. | See 4.3.2. |
| 9 | $I_{CES1}$ and $h_{FE1}$. | Not applicable. |
| 10 | 48 hours minimum. | 48 hours minimum. |
| 11 | $I_{CES1}$; $h_{FE1}$; $\Delta I_{CES1}$ = 100 percent of initial value or 5 nA dc, whichever is greater; $\Delta h_{FE1}$ = ±15 percent. | $I_{CES1}$ and $h_{FE1}$. |
| 12 | See 4.3.1. | See 4.3.1. |
| 13 | Subgroups 2 and 3 of table I herein; $\Delta I_{CES1}$ = 100 percent of initial value or 5 nA dc, whichever is greater; $\Delta h_{FE1}$ = ±15 percent. | Subgroup 2 of table I herein; $\Delta I_{CES1}$ = 100 percent of initial value or 5 nA dc, whichever is greater; $\Delta h_{FE1}$ = ±15 percent. |

(1) Shall be performed anytime after temperature cycling, screen 3a; JANTX and JANTXV levels do not need to be repeated in screening requirements.

4.3.1 Power burn-in conditions. Power burn-in conditions are shall be as follows: $V_{CB}$ = 10 - 30 V dc. Power shall be applied to achieve $T_J$ = +135°C minimum using a minimum $P_D$ = 75 percent of $P_T$ maximum, $T_A$ ambient rated as defined in 1.3. With approval of the qualifying activity and preparing activity, alternate burn-in criteria (hours, bias conditions, $T_J$, and mounting conditions) may be used for JANTX and JANTXV quality levels. A justification demonstrating equivalence is required. In addition, the manufacturing site's burn-in data and performance history will be essential criteria for burn-in modification approval. Use method 3100 of MIL-STD-750 to measure $T_J$.

4.3.2 Thermal impedance measurements. The thermal impedance measurements shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining $I_M$, $I_H$, $t_H$, $t_{MD}$ (and $V_C$ where appropriate). Measurement delay time ($t_{MD}$) = 70 µs max. See table III, subgroup 4 herein.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein. If alternate screening is being performed in accordance with MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of group A1 and A2 inspection only (table E-VIB, group B, subgroup 1 is not required to be performed again if group B has already been satisfied in accordance with 4.4.2).

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table II herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with 4.4.2.1 for quality level JANS and 4.4.2.2 for quality levels JAN, JANTX, and JANTXV.



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4.4.2.1 Quality level JANS. Group B inspection for quality level JANS shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIA of MIL-PRF-19500 and herein.

| Subgroup | Method | Condition |
|---|---|---|
| B4 | 1037 | $V_{CB}$ = 10 V dc, 2,000 cycles, adjust device current, or power, to achieve a minimum $\Delta T_J$ of +100°C. |
| B5 | 1027 | NOTE: If a failure occurs, resubmission shall be at the test conditions of the original sample. $V_{CB}$ = 10 V dc, $P_D$ ≥ 100 percent of maximum rated $P_T$ (see 1.3).<br><br>Option 1: 96 hours minimum, sample size in accordance with MIL-PRF-19500, table E-VIA, adjust $T_A$ or $P_D$ to achieve $T_J$ = +275°C minimum.<br><br>Option 2: 216 hours minimum, sample size = 45, c = 0; adjust $T_A$ or $P_D$ to achieve $T_J$ = +225°C minimum. |
| B5 | 2037 | Test condition D. |
| B6 | | Not applicable. |

4.4.2.2 Quality levels JAN, JANTX, and JANTXV. Group B inspection for quality levels JAN, JANTX, and JANTXV shall be conducted in accordance with the conditions specified in table E-VIC (small die flow) of MIL-PRF-19500 and herein. All catastrophic failures during CI shall be analyzed to the extent possible to identify root cause and corrective action. Whenever a failure is identified as wafer lot or wafer processing related, the entire wafer lot and related devices assembled from the wafer lot shall be rejected unless an appropriate determined corrective action to eliminate the failure mode has been implemented and the devices from the wafer lot are screened to eliminate the failure mode.

| Step | Method | Condition |
|---|---|---|
| 1 | 1026 | Steady-state life: 1,000 hours minimum, $V_{CB}$ = 10 V dc, power shall be applied to achieve $T_J$ = +150°C minimum using a minimum of $P_D$ = 75 percent of maximum rated $P_T$ as defined in 1.3. n = 45 devices, c = 0. The sample size may be increased and the test time decreased so long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours. |
| 2 | 1048 | Blocking life: $T_A$ = +150°C, $V_{CB}$ = 80 percent rated voltage, without exceeding max rated $V_{CEO}$. 48 hours minimum. n = 45 devices, c = 0. |
| 3 | 1032 | High-temperature life (non-operating), t = 340 hours, $T_A$ = +200°C. n = 22, c = 0. |

4.4.2.2.1 Sample selection. Samples selected for small die flow group B inspection shall be in accordance with all of the following requirements:

a. Samples shall be selected from an inspection lot that has been submitted to and passed table I, subgroup 2, conformance inspection.

b. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (step 1) may be tested prior to the application of final lead finish.

c. Separate samples may be used for each step.



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4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table E-VII of MIL-PRF-19500, and 4.4.3.1 for quality level JANS and 4.4.3.2 for quality levels JAN, JANTX, and JANTXV.

4.4.3.1 Quality level JANS.

| Subgroup | Method | Condition |
|---|---|---|
| C2 | 2036 | Test condition E; not applicable for UB devices. |
| C5 | 3131 | See 1.3. $R_{\theta JC}$ is applicable for all devices except UB. $R_{\theta JA}$ is applicable for UB devices. |
| C6 | 1026 | 1,000 hours, $V_{CB}$ = 10 V dc, power and ambient temperature shall be applied to the device to achieve $T_J$ = +150°C minimum, and minimum power dissipation of 75 percent of max rated $P_T$ (see 1.3 herein); n = 45, c = 0. The sample size may be increased and the test time decreased as long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours. |

4.4.3.2 Quality levels JAN, JANTX, and JANTXV.

| Subgroup | Method | Condition |
|---|---|---|
| C2 | 2036 | Test condition E; not applicable for UB devices. |
| C5 | 3131 | See 1.3, $R_{\theta JC}$ is applicable for all devices except UB. $R_{\theta JA}$ is applicable for UB devices. |
| C6 | | Not applicable. |

4.4.3.3 Sample selection. Samples for subgroups in group C shall be chosen at random from any inspection lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes table I tests herein for conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for C6 life test may be tested prior to the application of final lead finish. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

4.4.4 Group D inspection. Conformance inspection for radiation hardness assured JANS and JANTXV types shall include the group D tests specified in table II herein. These tests shall be performed as required in accordance with MIL-PRF-19500 and method 1019 of MIL-STD-750 for total ionizing dose, or method 1017 of MIL-STD-750 for neutron fluence as applicable (see 6.2.e herein), except group D, subgroup 2 may be performed separate from other subgroups. Alternate package options may also be substituted for the testing provided there is no adverse effect to the fluence profile.

4.4.5 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 as specified in table III herein.



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4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Collector-base time constant. This parameter may be determined by applying an rf signal voltage of 1.0 volt (rms) across the collector-base terminals, and measuring the ac voltage drop ($V_{eb}$) with a high- impedance rf voltmeter across the emitter-base terminals. With f = 79.8 MHz used for the 1.0 volt signal, the following computation applies:

$$r'_b , C_{c(ps)} = 2 X V_{eb} \text{ (millivolts)}$$

4.5.3 Displacement damage characterization. For RHA devices, each supplier shall perform a displacement damage characterization. The characterization shall demonstrate exposure versus data and does not indicate pass or fail criteria. The exposure shall be conducted in accordance with method 1017 of MIL-STD-750. The following details shall apply:

a. Samples may be taken from any wafer of the qualification lot.

b. As a minimum, testing shall be at 2E+12 n/cm2 plus two additional fluence levels chosen by the manufacturer.

c. If the device degrades less than 5 percent of the specification at the highest fluence level, a single data point may be sufficient.

Alternate package options may be substituted for characterization. The displacement damage characterization data shall be made available from the manufacturer.

4.6 Element evaluation of unencapsulated die. The element evaluation of unencapsulated die shall be in accordance with appendix G of MIL-PRF-19500. For subgroup 4, the burn-in duration for the JANKC level shall follows the JANS requirements and the JANHC shall follows the JANTX requirements.



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TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 2/ | | | | | | |
| Visual and mechanical examination | 2071 | | | | | |
| Solderability 3/ 4/ | 2026 | n = 15 leads, c = 0 | | | | |
| Resistance to solvents 3/ 4/ 5/ | 1022 | n = 15 devices, c = 0 | | | | |
| Salt atmosphere 6/ | 1041 | n = 6 devices, c = 0 | | | | |
| Temperature cycling (air to air) 3/ 4/ | 1051 | Test condition C, 25 cycles<br>n = 22 devices, c = 0 | | | | |
| Hermetic seal 4/<br>Fine and gross leak | 1071 | n = 22 devices, c = 0 | | | | |
| Electrical measurements 4/ | | Table I, subgroup 2 | | | | |
| Bond strength 3/ 4/ | 2037 | Precondition $T_A$ = +250°C at t = 24 hrs or $T_A$ = +300°C at t = 2 hrs,<br>n = 11 wires, c = 0 | | | | |
| Decap internal visual (design verification) 4/ | 2075 | n = 4 devices, c = 0 | | | | |
| Subgroup 2. | | | | | | |
| Thermal impedance 7/ | 3131 | See 4.3.2 | $Z_{\theta JX}$ | | | °C/W |
| Collector to base cutoff current | 3036 | Bias condition D; $V_{CB}$ = 140 V dc | $I_{CBO1}$ | | 10 | µA dc |
| Emitter to base cutoff current | 3061 | Bias condition D; $V_{EB}$ = 7 V dc | $I_{EBO1}$ | | 10 | µA dc |
| Collector to emitter breakdown voltage | 3011 | Bias condition D; $I_C$ = 30 mA dc; pulsed (see 4.5.1) | $V_{(BR)CEO}$ | 80 | | V dc |
| Collector to emitter cutoff current | 3041 | Bias condition C; $V_{CE}$ = 90 V dc | $I_{CES1}$ | | 10 | nA dc |
| Emitter to base cutoff current | 3061 | Bias condition D; $V_{EB}$ = 5 V dc | $I_{EBO2}$ | | 10 | nA dc |
| Forward current transfer ratio | 3076 | $V_{CE}$ = 10 V dc; $I_C$ = 150 mA dc; pulsed (see 4.5.1) | $h_{FE1}$ | 100 | 300 | |

See footnotes at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 2 – continued. | | | | | | |
| Forward current transfer ratio | 3076 | $V_{CE}$ = 10 V dc; $I_C$ = 0.1 mA dc; pulsed (see 4.5.1) | $h_{FE2}$ | 50 | 300 | |
| Forward current transfer ratio | 3076 | $V_{CE}$ = 10 V dc; $I_C$ = 10 mA dc; pulsed (see 4.5.1) | $h_{FE3}$ | 90 | | |
| Forward current transfer ratio | 3076 | $V_{CE}$ = 10 V dc; $I_C$ = 500 mA dc; pulsed (see 4.5.1) | $h_{FE4}$ | 50 | 300 | |
| Forward current transfer ratio | 3076 | $V_{CE}$ = 10 V dc; $I_C$ = 1 A dc; pulsed (see 4.5.1) | $h_{FE5}$ | 15 | | |
| Collector to emitter voltage (saturated) | 3071 | $I_C$ = 150 mA dc; $I_B$ = 15 mA dc; pulsed (see 4.5.1) | $V_{CE(sat)1}$ | | 0.2 | V dc |
| Collector to emitter voltage (saturated) | 3071 | $I_C$ = 500 mA dc; $I_B$ = 50 mA dc; pulsed (see 4.5.1) | $V_{CE(sat)2}$ | | 0.5 | V dc |
| Base to emitter voltage (saturated) | 3066 | Test condition A; $I_C$ = 150 mA dc; $I_B$ = 15 mA dc; pulsed (see 4.5.1) | $V_{BE(sat)}$ | | 1.1 | V dc |
| Subgroup 3 | | | | | | |
| High-temperature operation | | $T_A$ = +150°C | | | | |
| Collector to emitter cutoff current | 3041 | Bias condition C; $V_{CE}$ = 90 V dc | $I_{CES2}$ | | 5 | µA dc |
| Low-temperature operation | | $T_A$ = -55°C | | | | |
| Forward current transfer ratio | 3076 | $V_{CE}$ = 10 V dc; $I_C$ = 150 mA dc; pulsed (see 4.5.1) | $h_{FE6}$ | 40 | | |
| Subgroup 4 | | | | | | |
| Small-signal short- circuit forward-current transfer ratio | 3206 | $V_{CE}$ = 5 V dc; $I_C$ = 1 mA dc; f = 1 kHz | $h_{fe}$ | 80 | 400 | |
| Magnitude of small- signal short-circuit forward-current transfer ratio | 3306 | $V_{CE}$ = 10 V dc; $I_C$ = 50 mA dc; f = 20 MHz | $\|h_{fe}\|$ | 5 | 20 | |
| Input capacitance (output open circuited) | 3240 | $V_{EB}$ = 0.5 V dc; $I_C$ = 0; 100 kHz ≤ f ≤ 1 MHz | $C_{ibo}$ | | 60 | pF |
| Open circuit output capacitance | 3236 | $V_{CB}$ = 10 V dc; $I_E$ = 0; 100 kHz ≤ f ≤ 1 MHz | $C_{obo}$ | | 12 | pF |

See footnotes at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 4 | | | | | | |
| Noise figure | 3246 | $V_{CE}$ = 10 V dc; $I_C$ = 100 µA dc; $R_g$ = 1 kΩ; power bandwidth = 200 Hz f = 1 kHz | NF | | 4 | dB |
| Collector to base time constant | | $V_{CB}$ = 10 V dc; $I_C$ = 10 mA dc; f = 79.8 MHz (see 4.5.2) | $r'_b$, $C_c$ | | 400 | ps |
| Pulse response | | See figure 22 | $t_{on}$ + $t_{off}$ | | 30 | ns |
| Subgroup 5 | | | | | | |
| Safe operating area (continuous dc) | 3051 | $T_C$ = +25°C; t = 10 ms, 1 cycle (See figure 23) | | | | |
| Test 1 | | $V_{CE}$ = 10 V dc | | | | |
| 2N3019, 2N3019S | | $I_C$ = 500 mA dc | | | | |
| 2N3057A, 2N3700, 2N3700UB | | $I_C$ = 180 mA dc | | | | |
| Test 2 | | $V_{CE}$ = 40 V dc | | | | |
| 2N3019, 2N3019S | | $I_C$ = 130 mA dc | | | | |
| 2N3057A, 2N3700, 2N3700UB | | $I_C$ = 50 mA dc | | | | |
| Test 3 | | $V_{CE}$ = 80 V dc | | | | |
| 2N3019, 2N3019S | | $I_C$ = 60 mA dc | | | | |
| 2N3057A, 2N3700, 2N3700UB | | $I_C$ = 30 mA dc | | | | |
| Electrical measurements | | See table I, subgroup 2 herein | | | | |
| Subgroups 6 and 7 | | | | | | |
| Not applicable | | | | | | |

1/ For sampling plan see MIL-PRF-19500.
2/ For resubmission of failed test in subgroup 1 of table I, double the sample size of the failed test or sequence of tests. A failure in table I, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
3/ Separate samples may be used.
4/ Not required for JANS devices.
5/ Not required for laser marked devices.
6/ Required only for laser marked devices. Not required for non-corrosion prone base metals.
7/ For end-point measurements, this test is required for the following subgroups:
Group B, subgroup 3, 4, and 5 (JANS).
Group B, step 1 (JANTX and JANTXV).
Group C, subgroup 2 and 6.



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TABLE II. Group D inspection.

| Inspection 1/ 2/ 3/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 4/ | | | | | | |
| Neutron irradiation | 1017 | Neutron exposure $V_{CES}$ = 0 V | | | | |
| Collector to base cutoff current | 3036 | Bias condition D; $V_{CB}$ = 140 V dc | $I_{CBO1}$ | | 20 | µA dc |
| Emitter to base cutoff current | 3061 | Bias condition D; $V_{EB}$ = 7 V dc | $I_{EBO1}$ | | 20 | µA dc |
| Collector to emitter breakdown voltage | 3011 | Bias condition D; $I_C$ = 30 mA dc pulsed (see 4.5.1) | $V_{(BR)CEO}$ | 80 | | V dc |
| Collector to emitter cutoff current | 3041 | Bias condition C; $V_{CE}$ = 90 V dc | $I_{CES1}$ | | 20 | nA dc |
| Emitter to base cutoff current | 3061 | Bias condition D; $V_{EB}$ = 5 V dc | $I_{EBO2}$ | | 20 | nA dc |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 10 V dc; $I_C$ = 150 mA dc | [$h_{FE1}$] 5/ | [50] | 300 | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 10 V dc; $I_C$ = 0.1 mA dc | [$h_{FE2}$] 5/ | [25] | 300 | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 10 V dc; $I_C$ = 10 mA dc | [$h_{FE3}$] 5/ | [45] | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 10 V dc; $I_C$ = 500 mA dc | [$h_{FE4}$] 5/ | [25] | 300 | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 10 V dc; $I_C$ = 1 A dc | [$h_{FE5}$] 5/ | [7.5] | | |
| Collector-emitter saturation voltage | 3071 | $I_C$ = 150 mA dc; $I_B$ = 15 mA dc (see 4.5.1) | $V_{CE(sat)1}$ | | .23 | V dc |
| Collector-emitter saturation voltage | 3071 | $I_C$ = 500 mA dc; $I_B$ = 50 mA dc (see 4.5.1) | $V_{CE(sat)2}$ | | .58 | V dc |
| Base to emitter voltage (saturated) | 3066 | Test condition A; $I_C$ = 150 mA dc; $I_B$ = 15 mA dc; pulsed (see 4.5.1) | $V_{BE(sat)}$ | | 1.27 | V dc |

See footnotes at end of table.



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TABLE II. Group D inspection - Continued.

| Inspection 1/ 2/ 3/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 2 | | | | | | |
| * Total dose irradiation | 1019 | Condition A, Gamma exposure $V_{CES}$ = 64 V | | | | |
| Collector to base cutoff current | 3036 | Bias condition D; $V_{CB}$ = 140 V dc | $I_{CBO1}$ | | 20 | µA dc |
| Emitter to base cutoff current | 3061 | Bias condition D; $V_{EB}$ = 7 V dc | $I_{EBO1}$ | | 20 | µA dc |
| Collector to emitter breakdown voltage | 3011 | Bias condition D; $I_C$ = 30 mA dc pulsed (see 4.5.1) | $V_{(BR)CEO}$ | 80 | | V dc |
| Collector to emitter cutoff current | 3041 | Bias condition C; $V_{CE}$ = 90 V dc | $I_{CES1}$ | | 20 | nA dc |
| Emitter to base cutoff current | 3061 | Bias condition D; $V_{EB}$ = 5 V dc | $I_{EBO2}$ | | 20 | nA dc |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 10 V dc; $I_C$ = 150 mA dc | [$h_{FE1}$] 5/ | [50] | 300 | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 10 V dc; $I_C$ = 0.1 mA dc | [$h_{FE2}$] 5/ | [25] | 300 | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 10 V dc; $I_C$ = 10 mA dc | [$h_{FE3}$] 5/ | [45] | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 10 V dc; $I_C$ = 500 mA dc | [$h_{FE4}$] 5/ | [25] | 300 | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 10 V dc; $I_C$ = 1 A dc | [$h_{FE5}$] 5/ | [7.5] | | |
| Collector-emitter saturation voltage | 3071 | $I_C$ = 150 mA dc; $I_B$ = 15 mA dc; (see 4.5.1) | $V_{CE(sat)1}$ | | .23 | V dc |
| Collector-emitter saturation voltage | 3071 | $I_C$ = 500 mA dc; $I_B$ = 50 mA dc; (see 4.5.1) | $V_{CE(sat)2}$ | | .58 | V dc |
| Base to emitter voltage (saturated) | 3066 | Test condition A; $I_C$ = 150 mA dc; $I_B$ = 15 mA dc; pulsed (see 4.5.1) | $V_{BE(sat)}$ | | 1.27 | V dc |

1/ Tests to be performed on all devices receiving radiation exposure.
2/ For sampling plan, see MIL-PRF-19500.
3/ Electrical characteristics apply to the corresponding UB, suffix versions unless otherwise noted.
4/ See 6.2.e herein.
5/ See method 1019 of MIL-STD-750 for how to determine [$h_{FE}$] by first calculating the delta ($1/h_{FE}$) from the pre- and post-radiation $h_{FE}$. Notice the [$h_{FE}$] is not the same as $h_{FE}$ and cannot be measured directly. The [$h_{FE}$] value can never exceed the pre-radiation minimum $h_{FE}$ that it is based upon.



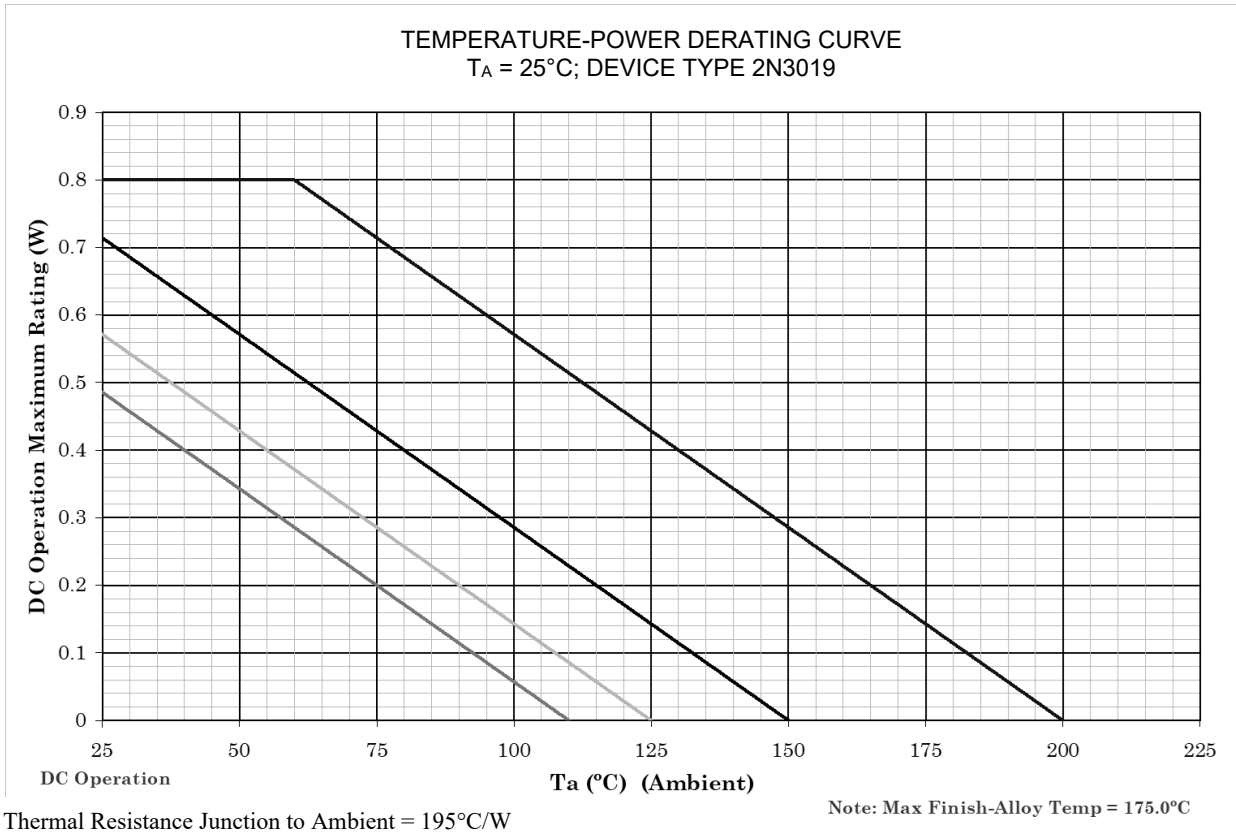
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TABLE III. Group E inspection (all quality levels) - for qualification and re-qualification only.

| Inspection | MIL-STD-750 | | Sample plan |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | 45 devices<br>c = 0 |
| Temperature cycling (air to air) | 1051 | Test condition C, 500 cycles. | |
| Hermetic seal<br>Fine leak<br>Gross leak | 1071 | | |
| Electrical measurements | | See table I, subgroup 2 herein. | |
| Subgroup 2 | | | 45 devices<br>c = 0 |
| Intermittent life | 1037 | $V_{CB}$ = 10 V dc, 6,000 cycles. | |
| Electrical measurements | | See table I, subgroup 2 herein. | |
| Subgroup 4 | | | 15 devices,<br>c = 0 |
| Thermal resistance | 3131 | $R_{\theta JSP(IS)}$ can be calculated but shall be measured once in the same package with a similar die size to confirm calculations (may apply to multiple specification sheets).<br>$R_{\theta JSP(AM)}$ need be calculated only. | |
| Thermal impedance curves | | See table E-IX of MIL-PRF-19500, subgroup 4. | Sample size<br>N/A |
| Subgroup 5 | | | |
| Not applicable | | | |
| Subgroup 6 | | | |
| ESD | 1020 | | |
| Subgroup 8 | | | 45 devices<br>c = 0 |
| Reverse stability | 1033 | Condition B. | |
| Subgroup 12 | | | |
| Neutron irradiation | 1017 | See 4.5.3. | |



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Thermal Resistance Junction to Ambient = 195°C/W

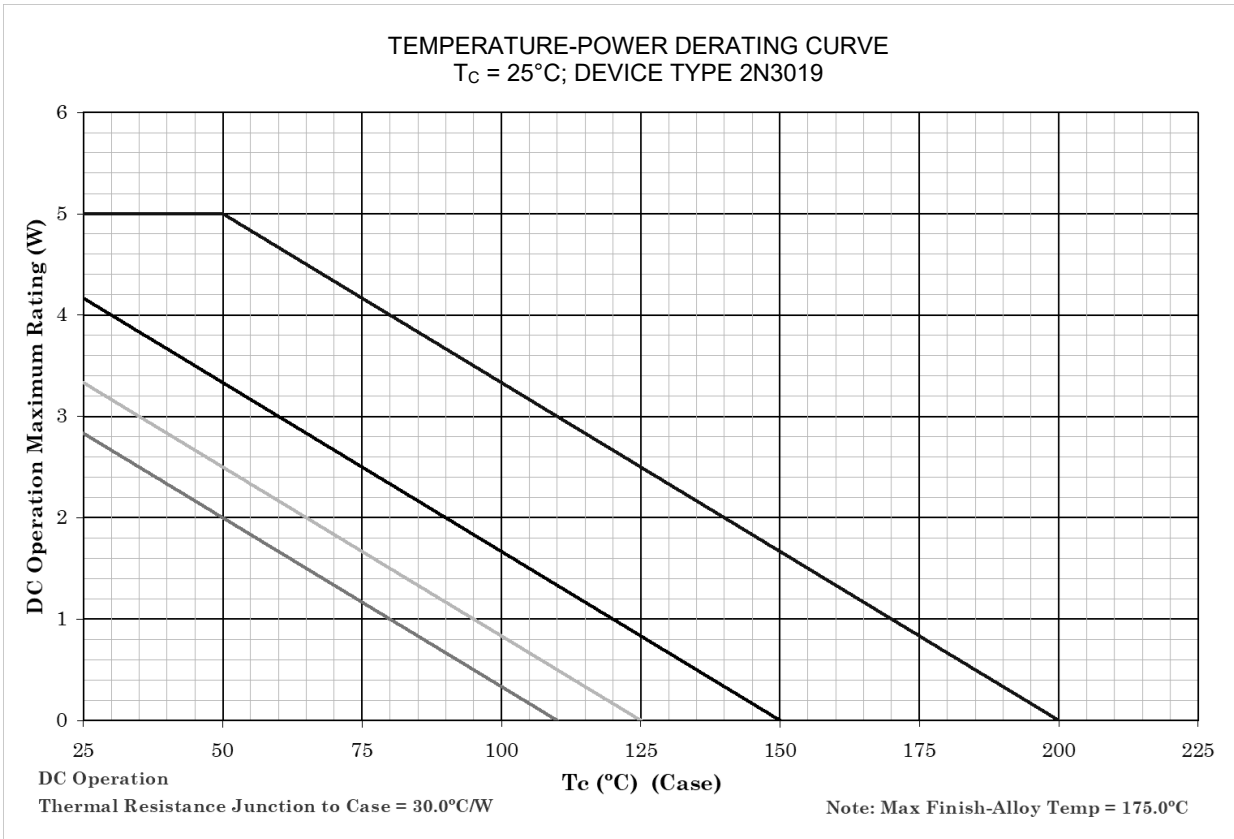
NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ $T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum $T_J$ allowed.
2. Derate design curve constrained by the maximum junction temperatures and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J$ ≤ +150°C, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at $T_J$ ≤ +125°C, and +110°C to show power rating where most users want to limit $T_J$ in their application.

* FIGURE 9. Derating for 2N3019 ($R_{\theta JA}$) PCB (TO-5 and TO-39).



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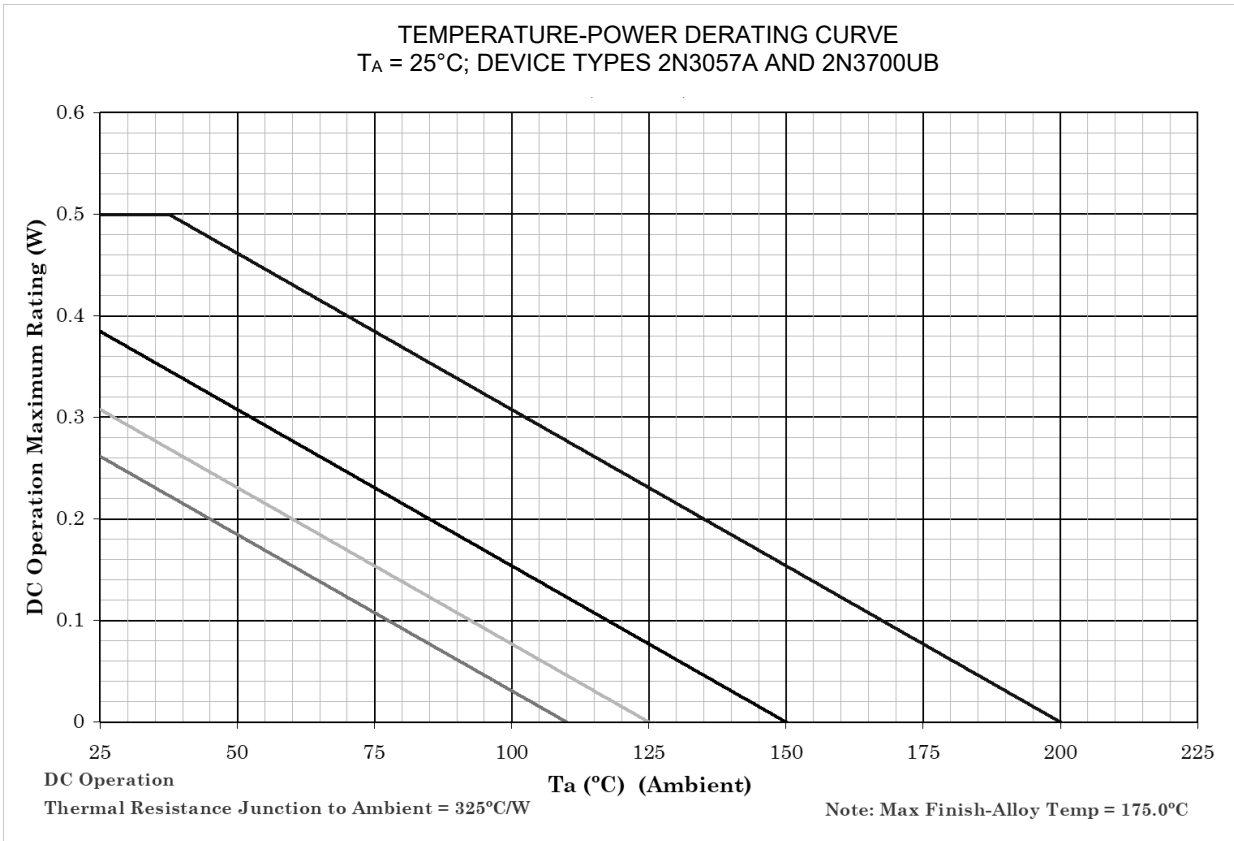

NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ $T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum $T_J$ allowed.
2. Derate design curve constrained by the maximum junction temperatures and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J$ ≤ +150°C, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at $T_J$ ≤ +125°C, and +110°C to show power rating where most users want to limit $T_J$ in their application.

* FIGURE 10. Derating for 2N3019 ($R_{\theta JC}$), base case mounted (TO-5 and TO-39).



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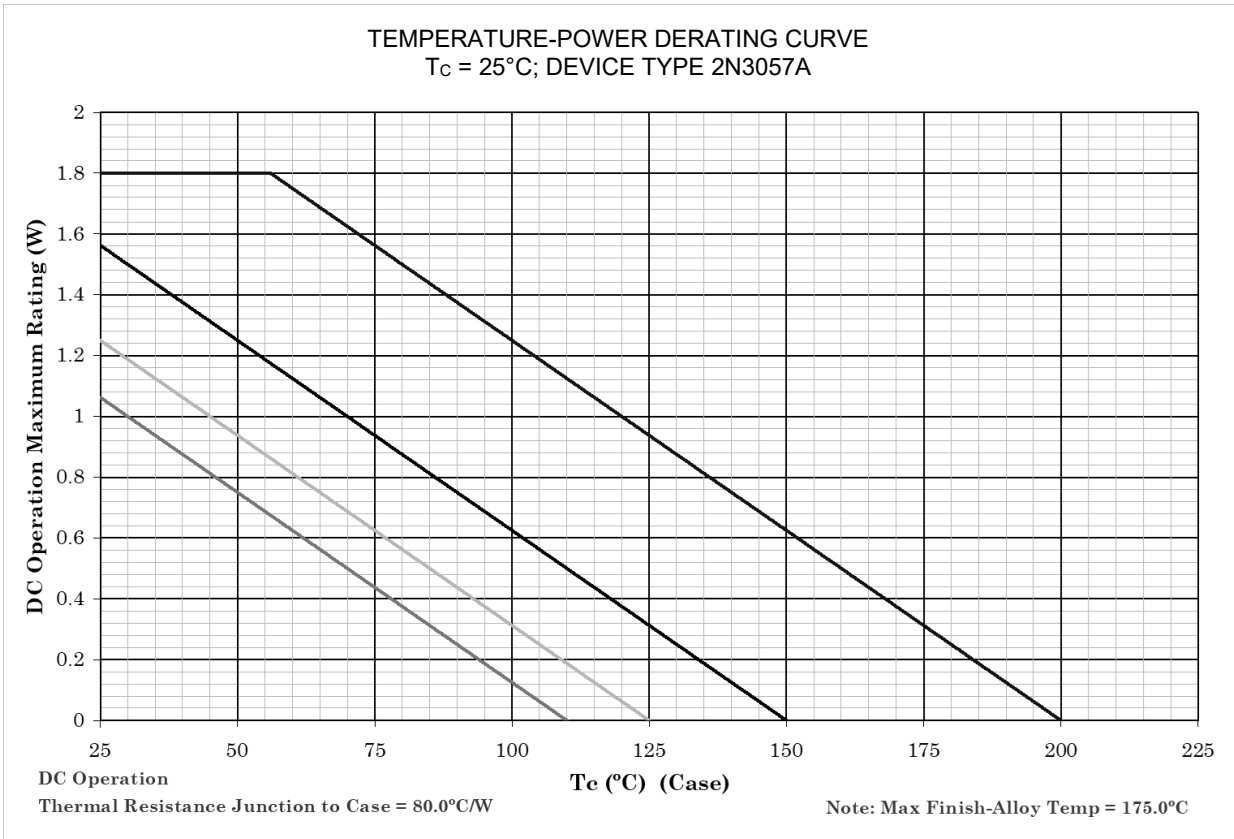

NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ $T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum $T_J$ allowed.
2. Derate design curve constrained by the maximum junction temperatures and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J$ ≤ +150°C, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at $T_J$ ≤ +125°C, and +110°C to show power rating where most users want to limit $T_J$ in their application.

* FIGURE 11. Derating for 2N3057A, 2N3700, and 2N3700UB ($R_{\theta JA}$) leads .125 inch (3.175 mm).



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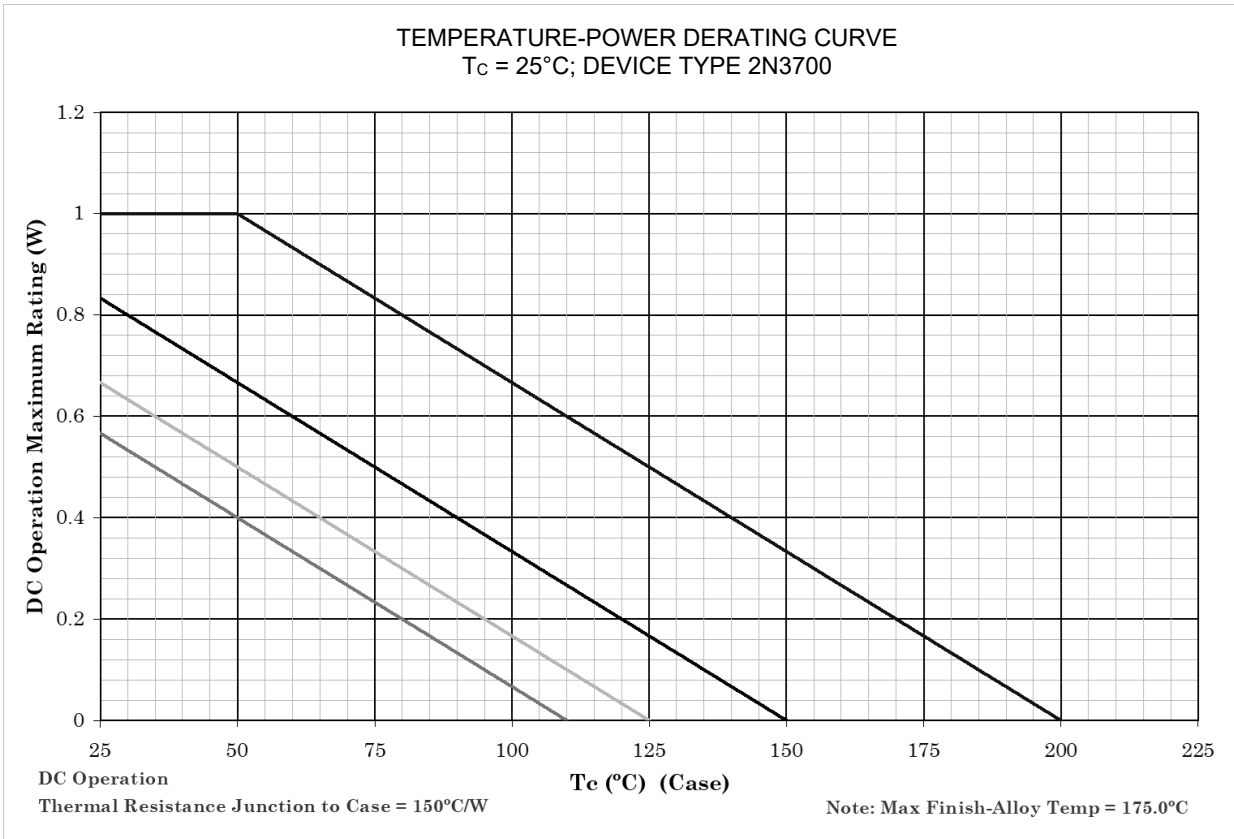

NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ $T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum $T_J$ allowed.
2. Derate design curve constrained by the maximum junction temperatures and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J$ ≤ +150°C, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at $T_J$ ≤ +125°C, and +110°C to show power rating where most users want to limit $T_J$ in their application.

* FIGURE 12. Derating for 2N3057A ($R_{\theta JC}$) (TO-46), base case mounted.



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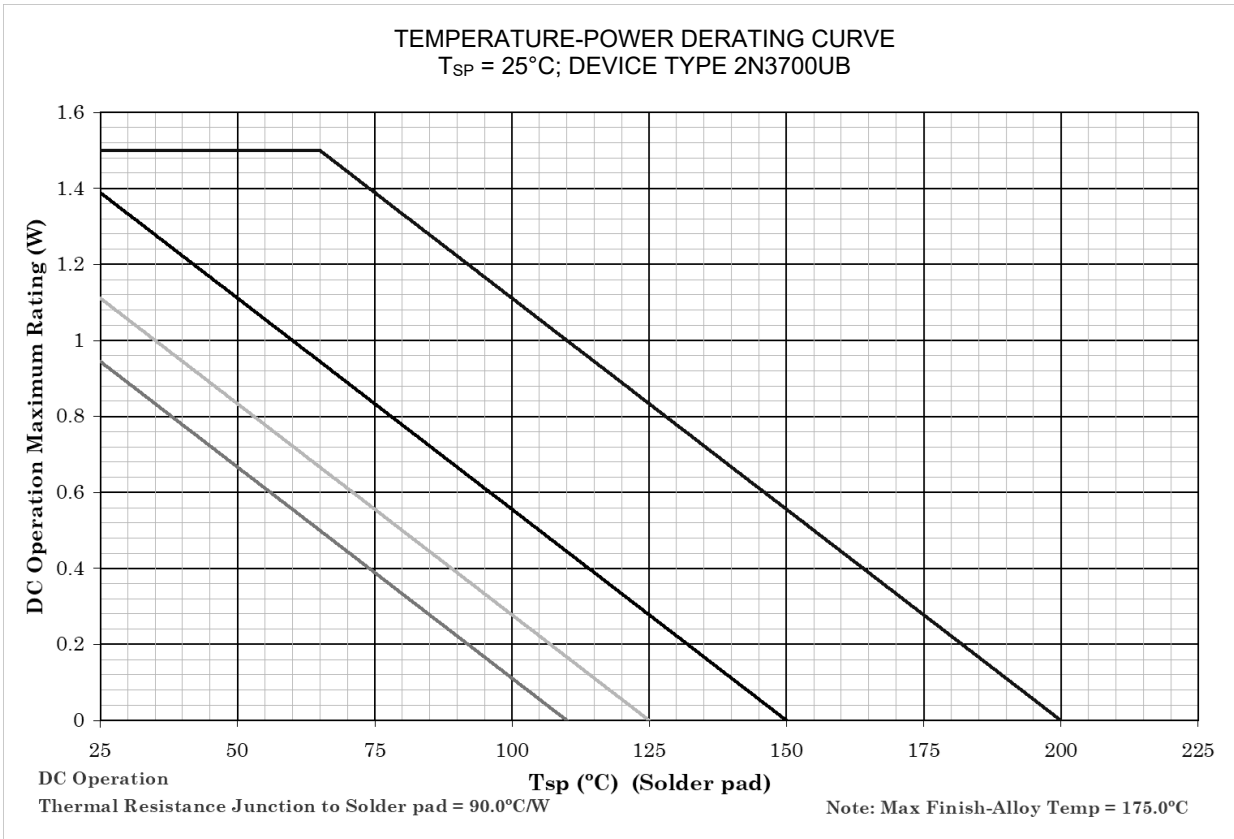

NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ $T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum $T_J$ allowed.
2. Derate design curve constrained by the maximum junction temperatures and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J$ ≤ +150°C, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at $T_J$ ≤ +125°C, and +110°C to show power rating where most users want to limit $T_J$ in their application.

* FIGURE 13. Derating for 2N3700 ($R_{\theta JC}$) (TO-18), base case mounted.



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NOTES:

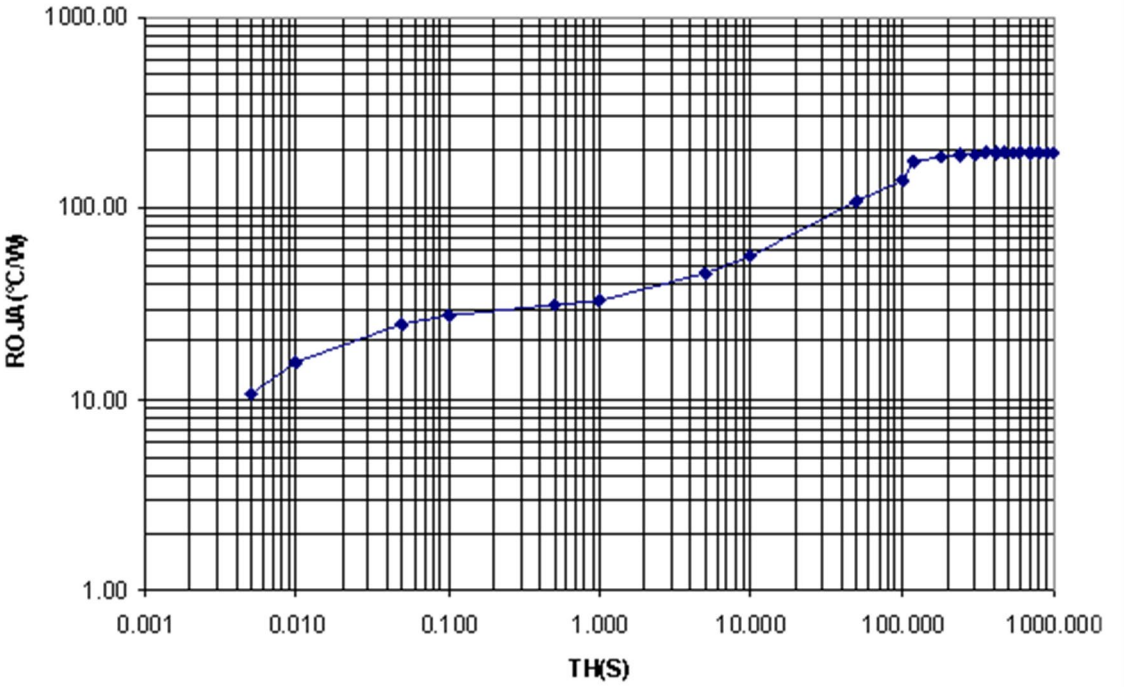
1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ $T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum $T_J$ allowed.
2. Derate design curve constrained by the maximum junction temperatures and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J$ ≤ +150°C, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at $T_J$ ≤ +125°C, and +110°C to show power rating where most users want to limit $T_J$ in their application.

* FIGURE 14. Derating for 2N3700UB ($R_{\theta JSP(IS)}$), infinite sink 3-points.



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MAXIMUM THERMAL IMPEDANCE
DEVICE TYPE 2N3019; $R_{\theta JA}$ FOR TO-205AA OR TO-205AD PACKAGE



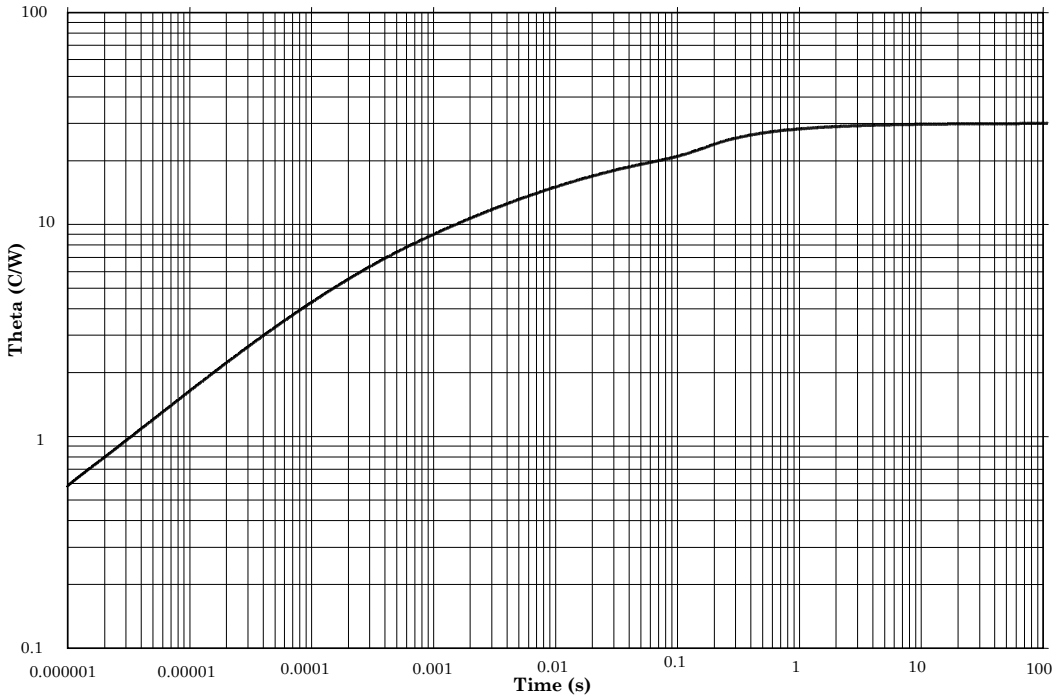

$R_{\theta JA}$ = 195°C/W

* FIGURE 15. Thermal impedance graph ($R_{\theta JA}$) for a 2N3019 in a TO-205AA or TO-205AD package.



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MAXIMUM THERMAL IMPEDANCE
DEVICE TYPE 2N3019;
$R_{\theta JC}$ FOR TO-205AA OR TO-205AD PACKAGE



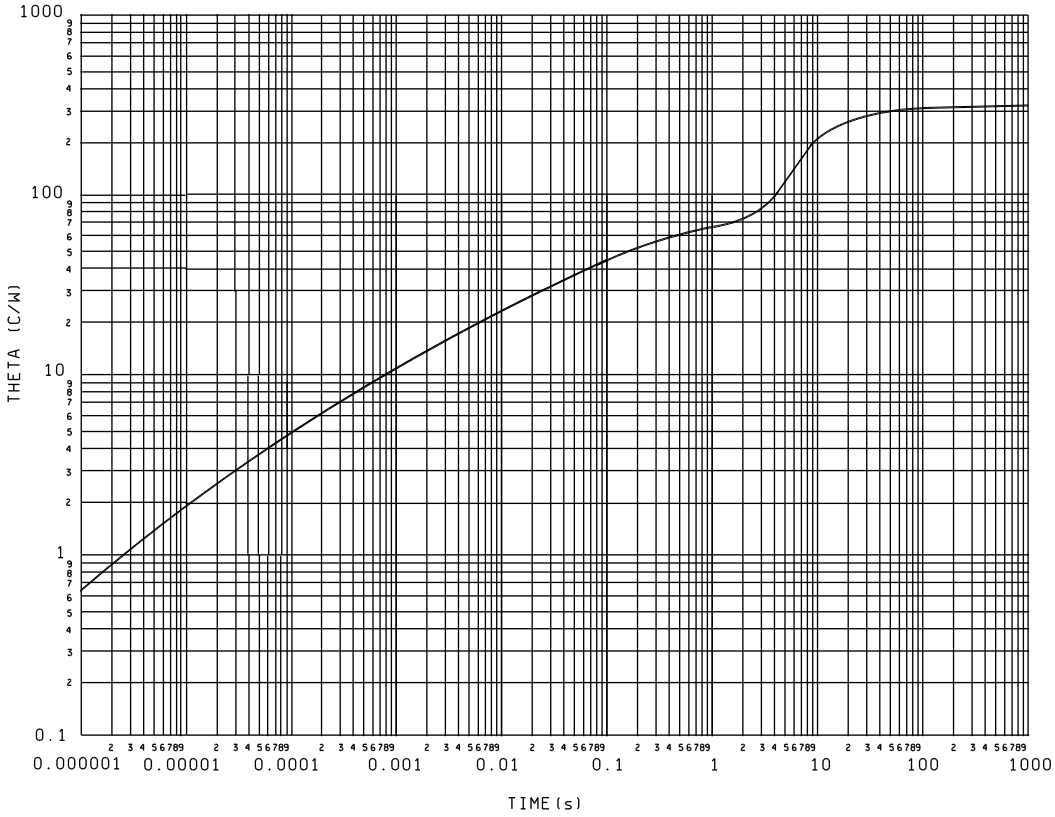

$R_{\theta JC}$ = 30°C/W

* FIGURE 16. Thermal impedance graph ($R_{\theta JC}$) for a 2N3019 in a TO-205AA or TO-205AD package.



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MAXIMUM THERMAL IMPEDANCE
DEVICE TYPE 2N3057A;
$R_{\theta JA}$ FOR TO-206AB PACKAGE



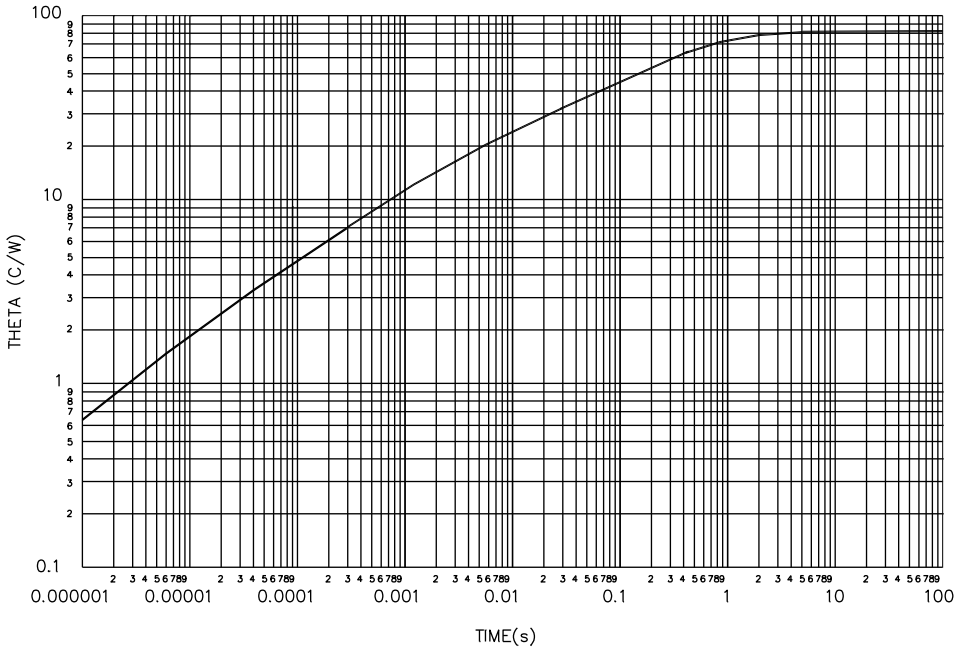

$R_{\theta JA}$ = 325°C/W

* FIGURE 17. Thermal impedance graph ($R_{\theta JA}$) for devices (2N3057A) in TO-206AB package.



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MAXIMUM THERMAL IMPEDANCE
DEVICE TYPE 2N3057A;
$R_{\theta JC}$ FOR TO-206AB PACKAGE



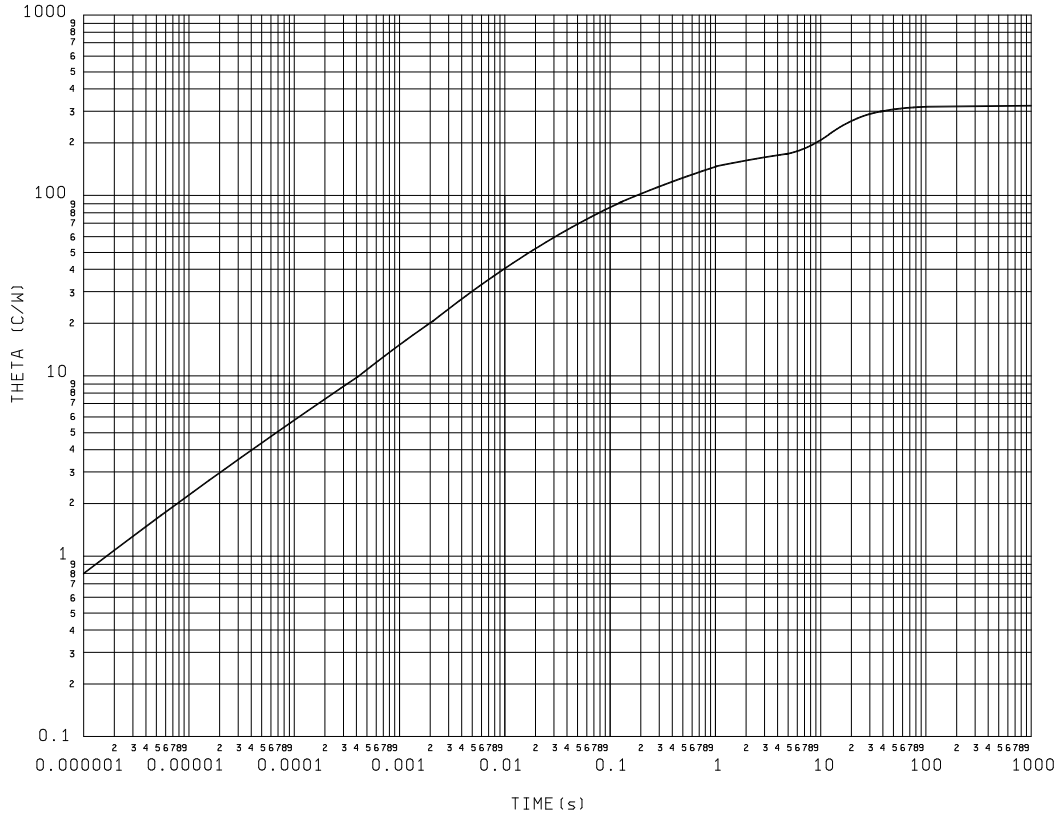

$R_{\theta JC}$ = 80°C/W

* FIGURE 18. Thermal impedance graph ($R_{\theta JC}$) for devices (2N3057A) in TO-206AB package.



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MAXIMUM THERMAL IMPEDANCE
DEVICE TYPE 2N3700;
$R_{\theta JA}$ FOR TO-206AA PACKAGE



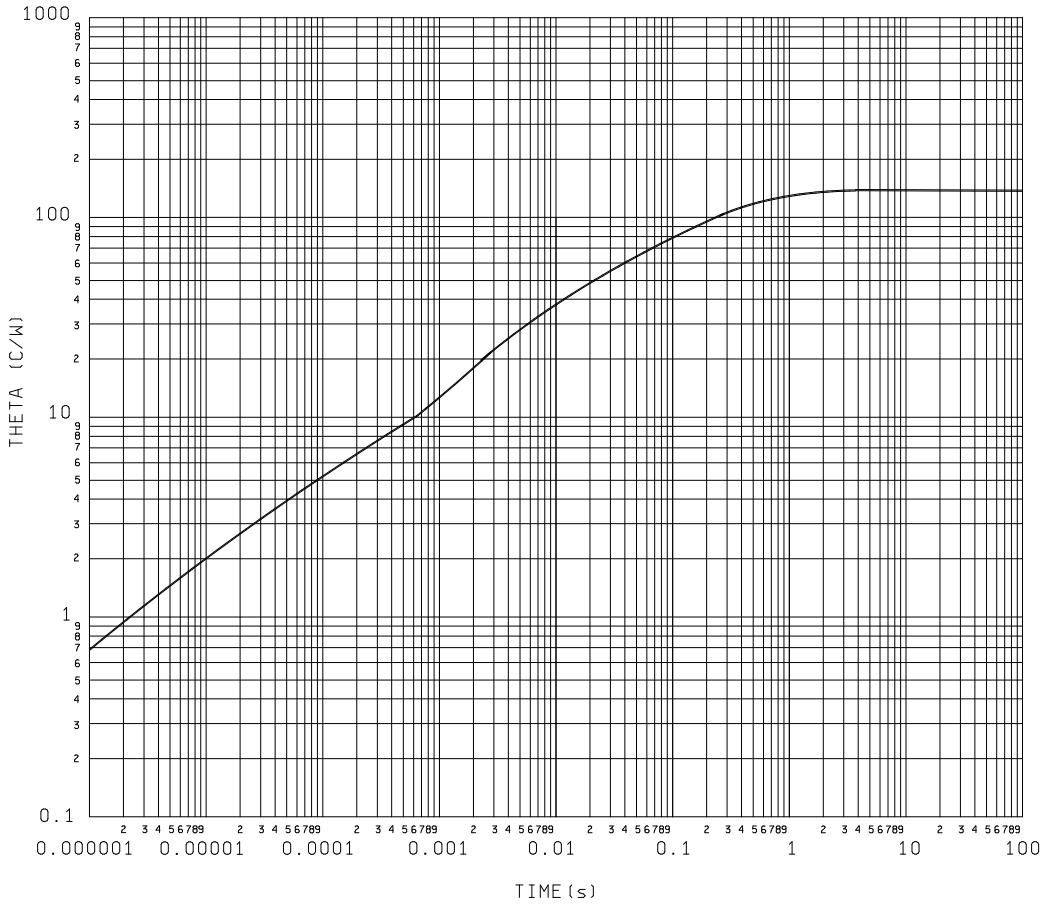

$R_{\theta JA}$ = 325°C/W

FIGURE 19. Thermal impedance graph ($R_{\theta JA}$) for devices (2N3700) in a TO-206AA package.



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Maximum Thermal Impedance
DEVICE TYPE 2N3700;
$R_{\theta JC}$ FOR TO-206AA PACKAGE



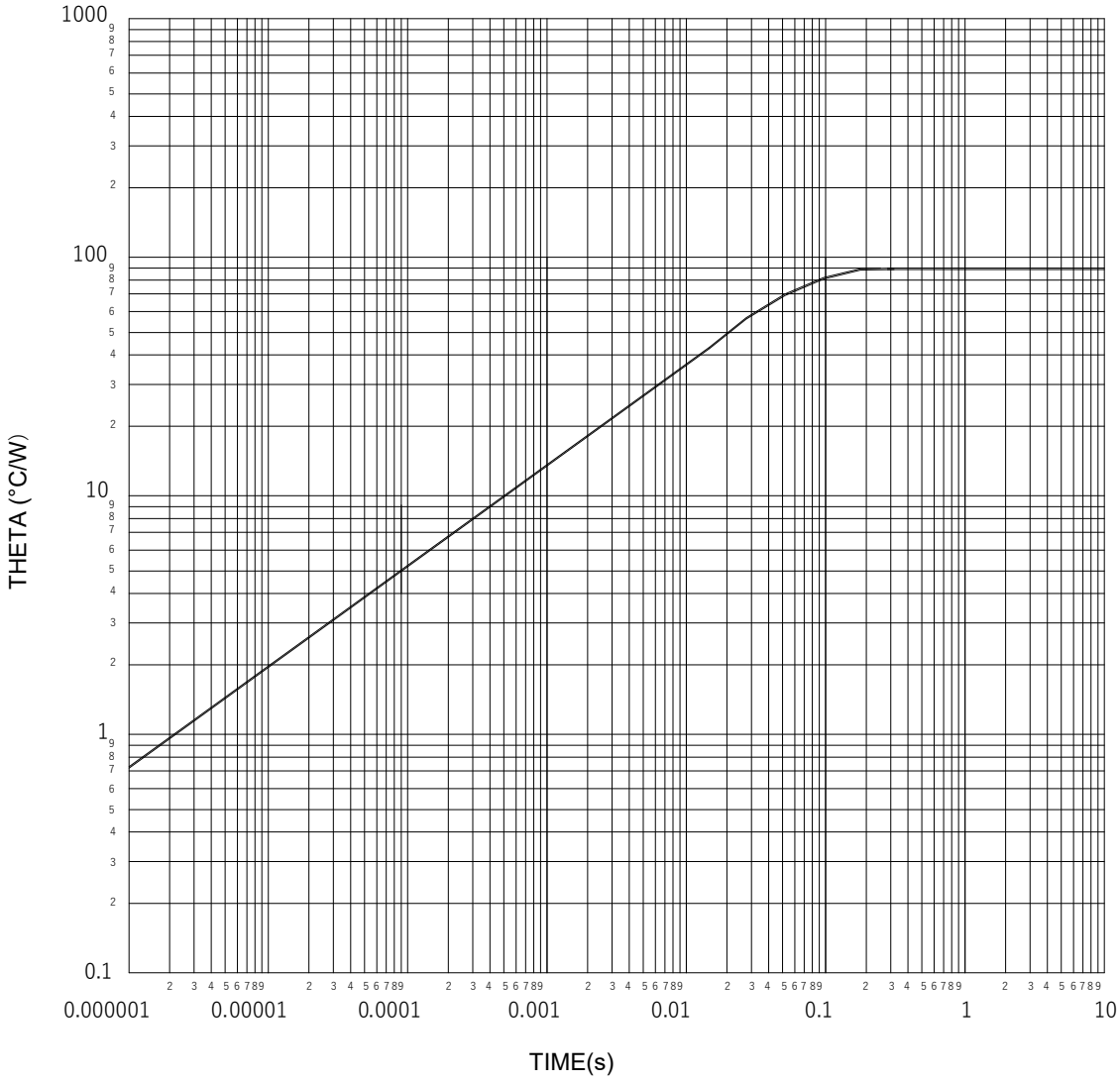

$R_{\theta JC}$ = 150°C/W

* FIGURE 20. Thermal impedance graph ($R_{\theta JC}$) for devices (2N3700) in a TO-206AA package.



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MAXIMUM THERMAL IMPEDANCE
DEVICE TYPE 2N3700;
$R_{\theta JSP(IS)}$ FOR UB PACKAGE

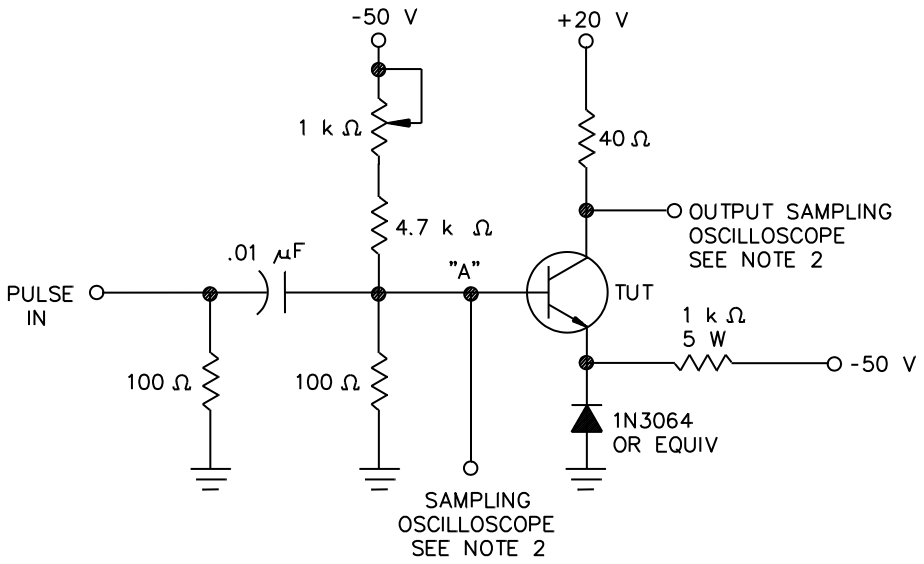


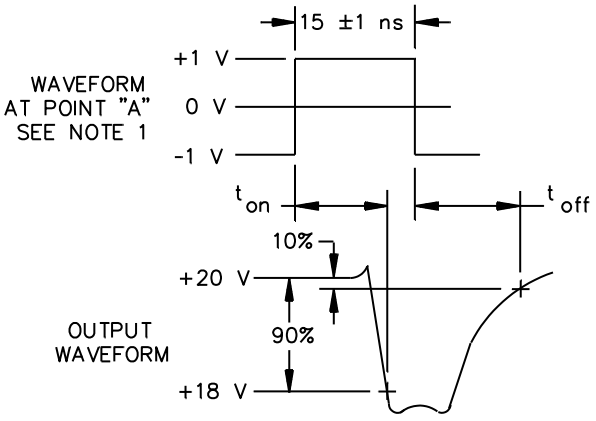

$R_{\theta JSP(IS)}$ = 90°C/W

* FIGURE 21. Thermal impedance graph ($R_{\theta JSP(IS)}$) for a 2N3700 in a UB package.



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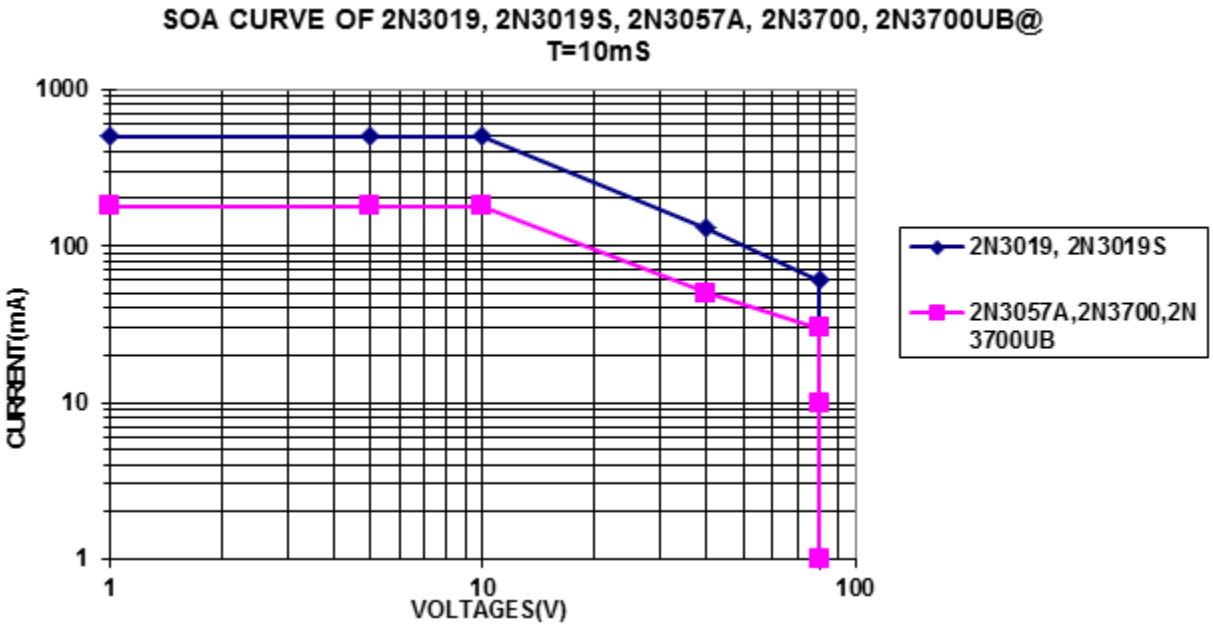

NOTES:

1. The rise time ($t_r$) of the applied pulse shall be ≤ 2.0 ns, duty cycle ≤ 2 percent and the generator source impedance shall be 50 ohms.
2. Sampling oscilloscope: $Z_{IN}$ ≥ 100 kΩ, $C_{IN}$ ≤ 12 pF, rise time ≤ 2.0 ns.

* FIGURE 22. Non-saturated switching-time test circuit.



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* FIGURE 23. Maximum safe operating graph (10 ms).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.



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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

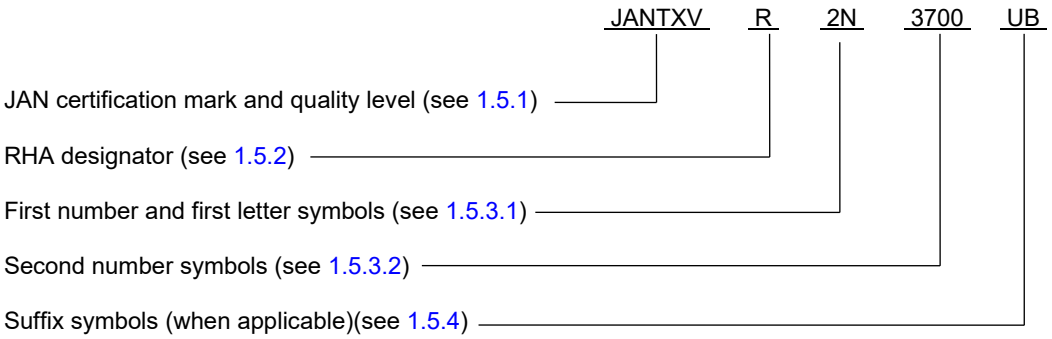
d. The complete PIN, see 1.5 and 6.4.

e. For acquisition of RHA designed devices, table II, subgroup 1 testing of group D is optional. If table II, subgroup 1 testing is desired, it must be specified in the contract.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

6.4 PIN construction examples.

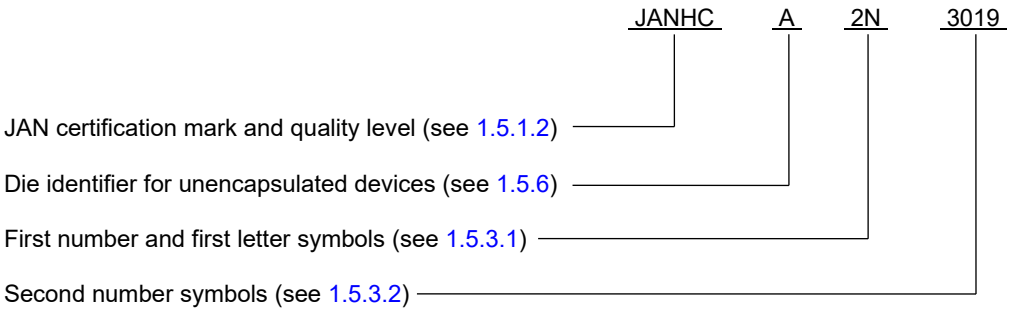
6.4.1 Encapsulated devices The PINs for encapsulated devices are constructed using the following form.






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6.4.2 Un-encapsulated devices. The PINs for un-encapsulated devices are constructed using the following form.




6.5 List of PINs.

6.5.1 PINs for encapsulated devices. The following is a list of possible PINs for encapsulated devices available on this specification sheet.

| PINs for types 2N3019 (1) | PINs for types 2N3019S (1) | PINs for types 2N3057A (1) | PINs for types 2N3700 (1) | PINs for types 2N3700UB (1) |
|---|---|---|---|---|
| JAN2N3019 | JAN2N3019S | JAN2N3057A | JAN2N3700 | JAN2N3700UB |
| JANTX2N3019 | JANTX2N3019S | JANTX2N3057A | JANTX2N3700 | JANTX2N3700UB |
| JANTXV2N3019 | JANTXV2N3019S | JANTXV2N3057A | JANTXV2N3700 | JANTXV2N3700UB |
| JANS2N3019 | JANS2N3019S | JANS2N3057A | JANS2N3700 | JANS2N3700UB |
| JANTXVR2N3019 | JANTXVR2N3019S | JANTXVR2N3057A | JANTXVR2N3700 | JANTXVR2N3700UB |
| JANTXVF2N3019 | JANTXVF2N3019S | JANTXVF2N3057A | JANTXVF2N3700 | JANTXVF2N3700UB |
| JANS#2N3019 | JANS#2N3019S | JANS#2N3057A | JANS#2N3700 | JANS#2N3700UB |

(1) The number sign (#) represents one of eight RHA designators available (M, D, P, L, R, F, G, or H).

* 6.5.2 PINs for unencapsulated die. The following is a list of possible PINs for unencapsulated die available on this specification sheet.

| PINs for types 2N3019 (1) | | PINs for types 2N3700 (1) | | | |
|---|---|---|---|---|---|
| JANHCA2N3019 | JANHCD2N3019 | JANHCA2N3700 | JANHCB2N3700 | JANHCC2N3700 | JANHCD2N3700 |
| JANKCA2N3019 | JANKCD2N3019 | JANKCA2N3700 | JANKCB2N3700 | JANKCC2N3700 | JANKCD2N3700 |
| JANHCA#2N3019 | JANHCD#2N3019 | JANHCA#2N3700 | JANHCB#2N3700 | JANHCC#2N3700 | JANHCD#2N3700 |
| JANKCA#2N3019 | JANKCD#2N3019 | JANKCA#2N3700 | JANKCB#2N3700 | JANKCC#2N3700 | JANKCD#2N3700 |

(1) The number sign (#) represents one of eight RHA designators available (M, D, P, L, R, F, G, or H).



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* 6.6 Suppliers of JANHC and JANKC die. The qualified JANHC and JANKC suppliers with the applicable letter version (example JANHCA2N3700) will be identified on the QML.

| Die ordering information | | | | |
|---|---|---|---|---|
| PIN | Manufacturer | | | |
| | 34156 | 43611 | 52GC4 | |
| 2N3700 | JANHCA2N3700<br>JANKCA2N3700 | JANHCB2N3700<br>JANKCB2N3700 | JANHCC2N3700<br>JANKCC2N3700 | JANHCD2N3700<br>JANKCD2N3700 |
| 2N3019 | JANHCA2N3019<br>JANKCA2N3019 | | | JANHCD2N3019<br>JANKCD2N3019 |

6.7 Request for new types and configurations. Requests for new device types or configurations for inclusions in this specification sheet should be submitted to: DLA Land and Maritime, ATTN: VAC, Post Office Box 3990, Columbus, OH 43218-3990 or by electronic mail at Semiconductor@dla.mil or by facsimile (614) 693-1642 or DSN 850-6939.

* 6.8 Amendment notations. The margins of this specification are marked with asterisks to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:
Army - CR
* Navy - SH
Air Force - 85
NASA - NA
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, MC
Air Force - 19

Preparing activity:
DLA - CC

(Project 5961-2021-050)

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